

Expander properties of superspecial isogeny digraphs with level structure

Thomas Decru & Krijn Reijnders

KU Leuven

Abstract. Charles, Goren and Lauter proved that the supersingular ℓ -isogeny graph is a Ramanujan graph, which is an optimal expander. Jordan and Zaytman argued that this is no longer true in dimension two, but Florit and Smith showed that those graphs exhibit good expansion properties nonetheless. Castryck, Decru and Smith however have pointed out that the higher-dimensional analogue setting should only consider a subset of all edges, namely the paths corresponding to (ℓ^k, ℓ^k) -isogenies, so-called *good extensions*, instead of all $(\ell^a, \ell^b, \ell^c, \ell^d)$ -isogenies in general, which contain bad extensions too. Such bad extensions lead to many small cycles in the graph, which are a cryptographic problem due to collisions and a graph-theoretic nuisance as these superfluous edges counteract part of the expansion properties. Restricting to good extensions makes the resulting graph directed, as outgoing edges now depend on the incoming edge.

We study (ℓ, ℓ) -level surfaces and $(\ell)^g$ -isogeny digraphs restricted to good extensions for concrete small dimensions and degrees ℓ . These graphs exhibit excellent expander properties: by our heuristic evidence, they are Ramanujan graphs for all primes ℓ in dimension 1, and for $\ell = 2$ in dimension 2. Our main conjecture implies that this would still be the case for $\ell = 3$ in dimension 2, but not for any larger ℓ in dimension 2, or any ℓ in dimension 3 and up. Furthermore, we generalize the work of Florit and Smith from $\ell = 2$ to general primes ℓ , by classifying all abelian surfaces with nontrivial automorphism groups and their actions on their maximal isotropic (ℓ, ℓ) -subgroups.

1 Introduction

In 2006, Charles Goren and Lauter constructed cryptographic hash functions from expander graphs [CLG09]. One of their constructions relied on computing low-degree isogenies between supersingular elliptic curves over a field of cryptographically large characteristic. Using a trusted setup [BCCF+23, EHL+18, KLPT14], this hash function is still assumed to be secure and the underlying mathematical structure is used in various other cryptographic protocols

such as SQIsign [FKLP+20] and the - now fallen [CD23, MMPP+23, Rob23] - SIDH [JF11]. Such isogeny graphs are Ramanujan graphs, which are optimal expanders, and Charles, Goren and Lauter have shown in 2009 that they generalize to higher dimensions as well [CGL09]. Unfortunately, this set-up requires using abelian varieties with real multiplication, which is a nontrivial generalization of elliptic curves and a nuisance for cryptographic purposes due to the required overhead.

In 2018, Takashima [Tak18] proposed a leaner - and arguably more natural - version of the two-dimensional hash function, namely abelian surfaces *without* real multiplication, but it was swiftly broken by Flynn and Ti in 2019 [FT19]. Indeed: the composition of two (ℓ, ℓ) -isogenies does not necessarily lead to an (ℓ^2, ℓ^2) -isogeny, but can also compose to an (ℓ^2, ℓ, ℓ) -isogeny, leading to many small cycles in the graph, resulting in trivially-found collisions for the hash function. Castryck, Decru and Smith revisited the construction [CDS20], identifying all these cycles and giving a straightforward way to avoid them in all dimensions. In essence, the fix consists of labeling every visited vertex with a so-called level structure referring to the last-traversed edge, and only allowing the walk in the graph to continue along paths which take this into account, namely the *good extensions*. This resulting higher-dimensional hash function is still assumed to be secure, although it has recently been shown that it also requires a trusted setup [CDKL+25], just as in the one-dimensional case.

Using Kani’s lemma [Kan97], a lot of the focus in isogeny-based cryptography has shifted towards higher-dimensional variants, including - but not limited to - protocols such as PRISM [BBCS+25], higher-dimensional variants of SQIsign [BDFL+24, DF24, DLRW24, NOCC+24], PEGASIS [DEFM+25] and MIKE [Rob24]. Because of this, the expander properties of the underlying graphs in higher dimensions are important to take into account for the security of some of these new protocols. Jordan and Zaytman proved that the graphs constructed in Takashima’s hash function are no longer Ramanujan [JZ25], whereas Florit and Smith showed that they exhibit good expansion properties nonetheless [FS22]. Unfortunately, neither of these works have taken into account the fact that the small cycles should be avoided, and these indubitably weaken the expander properties. Furthermore, both works do not take into account that these graphs are, by definition, directed.

As an introductory example, consider $E_0 : y^2 = x^3 + 1$ with automorphism $\zeta^{(3)} : (x, y) \mapsto (\zeta_3 x, y)$. This automorphism acts on $E_0[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^2$ as an element of $\mathbb{F}_\ell^{2 \times 2}$ with minimal polynomial $x^2 + x + 1$. If $\ell \equiv 2 \pmod{3}$, this polynomial is irreducible so we can find a basis of $E_0[\ell]$ such that $\zeta^{(3)}$ can be represented as $\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. By Burnside’s lemma, it follows that there are only $(\ell + 1)/3$ distinct cyclic subgroups of order ℓ . In the corresponding ℓ -isogeny graph, this node thus only has $(\ell + 1)/3$ outgoing edges. If $\ell \equiv 1 \pmod{3}$ however, the automorphism will act as $\begin{pmatrix} \mu_3 & 0 \\ 0 & \mu_3^2 \end{pmatrix}$ on some choice of basis $E_0[\ell]$. We then have two one-dimensional $\zeta^{(3)}$ -invariant subspaces, leading to $(\ell + 5)/3$ cyclic subgroups of order ℓ .

Technical overview

To get a better understanding of the two-dimensional isogeny graph, we start by listing all possible numbers of outgoing edges of principally polarized abelian surfaces in this graph. Generically, there are $\ell^3 + \ell^2 + \ell + 1$ outgoing edges, one for each maximal isotropic subgroup $G \cong (\mathbb{Z}/\ell\mathbb{Z})^2 \subsetneq A[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^4$. Due to nontrivial automorphisms on A however, this is not always the case, which is exactly why the graph is impossible to make undirected.

An example. Consider the genus-2 curve $C/\overline{\mathbb{F}}_p : y^2 = x^5 - 1$ and let J be its Jacobian. The hyperelliptic involution $(x, y) \mapsto (x, -y)$ is an automorphism on C and induces the automorphism $[-1]$ on J . This automorphism exists on every abelian variety and does not impact the number of outgoing edges in the graph since it always maps maximal isotropic subgroups to itself. For this reason, we will work with the reduced automorphism group $\text{RA}(J) = \text{Aut}(J)/[-1]$. In this example, we have that $\text{RA}(J) = \mathbb{Z}/5\mathbb{Z}$ with generator $\zeta_*^{(5)}$, induced by the automorphism $(x, y) \mapsto (\zeta_5 x, y)$ on C , where ζ_5 is a fifth root of unity in $\overline{\mathbb{F}}_p$.

We are now interested in the action of $\zeta_*^{(5)}$ on pairs (J, G) , where $G \subsetneq J[\ell]$ is a maximal isotropic subgroup, where we assume $\ell \neq 5$ for the sake of simplicity. Since $\zeta_*^{(5)}$ has order five, its minimal polynomial is $\Phi_5 = x^4 + x^3 + x^2 + x + 1$.

If $\ell \equiv 1 \pmod{5}$, we have that $\Phi_5 \pmod{\ell}$ splits into linear factors, hence it must act as the diagonal 4×4 -matrix $\text{diag}(\mu_5, \mu_5^2, \mu_5^4, \mu_5^3) \in \text{Sp}_4(\mathbb{F}_\ell)$ on some symplectic choice of basis $\langle D_1, D_2, D'_1, D'_2 \rangle = J[\ell]$, where μ_5 is a fifth root of unity in $\overline{\mathbb{F}}_\ell$. One can then argue by eigenspace decomposition that the only $\zeta_*^{(5)}$ -invariant maximal isotropic subgroups are $\langle D_1, D_2 \rangle, \langle D_1, D'_2 \rangle, \langle D'_1, D_2 \rangle, \langle D'_1, D'_2 \rangle$. If $\ell \equiv 2, 3 \pmod{5}$, we have that $\Phi_5 \pmod{\ell}$ is irreducible and there are no $\zeta_*^{(5)}$ -invariant maximal isotropic subgroups of $J[\ell]$. If $\ell \equiv 4 \pmod{5}$, we have that Φ_5 factors as

$$\Phi_5 = (x^2 - (\mu_5 + \mu_5^4) \cdot x + 1) \cdot (x^2 - (\mu_5^2 + \mu_5^3) \cdot x + 1) \pmod{\ell},$$

and so the action of $\zeta_*^{(5)}$ on $J[\ell]$ can be seen as the 4×4 -matrix

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & \mu_5 + \mu_5^4 & 0 \\ 0 & 1 & 0 & \mu_5^2 + \mu_5^3 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell)$$

for some symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. This matrix consists of two “blocks” (rows/columns 1 and 3 and rows/columns 2 and 4), so it decomposes as a direct summand of subspaces. One can show that since the blocks are not similar, the only possible $\zeta_*^{(5)}$ -invariant maximal isotropic subgroups are $\langle D_1, D'_1 \rangle$ and $\langle D_2, D'_2 \rangle$, but these are not symplectic. By Burnside’s lemma, we can conclude that the number of (J, G) up to isomorphism for this example is given by

$$\begin{cases} \frac{1}{5}(\ell^3 + \ell^2 + \ell + 17) & \text{if } \ell \equiv 1 \pmod{5}, \\ \frac{1}{5}(\ell^3 + \ell^2 + \ell + 1) & \text{if } \ell \not\equiv 1 \pmod{5}. \end{cases}$$

This gives us the exact number of (ℓ, ℓ) -level surfaces for this specific Jacobian. In this work, we do the same for all other principally polarized abelian surfaces by a close analysis of their reduced automorphism groups. These results are summarized in [Table 1](#) for Jacobians of genus-2 curves and in [Table 2](#) for products of elliptic curves.

Isogeny Digraphs. Given the understanding of automorphisms on level structures, we may then explore the *superspecial* $(\ell)^g$ -isogeny digraph, whose vertices are isomorphism classes (A, H) , where A is a g -dimensional superspecial abelian variety, and $H \subseteq A[\ell]$ a maximal isotropic subgroup. Edges are derived from those $(\ell)^g$ -isogenies $\varphi : A \rightarrow A'$ whose kernels intersect trivially with H .

This directed graph accurately describes the behaviour of isogeny walks with no backtracking: every combination of two edges is a *good extension*, and a walk of length n corresponds to an $(\ell^n)^g$ -isogeny, making this digraph fundamentally different from the usual ℓ -isogeny graphs studied in isogeny-based cryptography. We posit that the expansion properties of this digraph more accurately reflect the behaviour of isogeny walks as used in isogeny-based cryptography.

Surprisingly, the concrete expansion properties of these digraphs has hardly been studied. We provide heuristic data for $g = 1$ for a range of small prime powers ℓ^k and for $g = 2$ for $\ell = 2$. This digraph is *Ramanujan*, implying optimal expansion properties, in contrast to previous results on the $(2, 2)$ -isogeny graph with backtracking. Based on these observations, we derive our [Main Conjecture](#). Informally, our conjecture states that the second eigenvalue λ_2 of the adjacency matrix of such graphs over increasingly larger fields $\mathbb{F}_{p^2}$ converges to

$$\lim_{p \rightarrow \infty} \lambda_2 = \ell^{g^2/2}.$$

This agrees with the research on expander properties for $g = 1$, and furthermore implies that these digraphs are Ramanujan for $\ell \in \{2, 3\}$ for $g = 2$. More concretely, our conjecture poses that taking only good extensions strongly improves the expansion properties of isogeny walks, which is a significant result for the fundamental security of higher-dimensional isogeny-based cryptography.

Our contributions

This work explores the action of automorphisms on (ℓ, ℓ) -level surfaces and the expansion properties of superspecial isogeny digraphs.

- (i) We catalog the action of automorphisms on (ℓ, ℓ) -level structure, both for Jacobians and elliptic products. This generalizes previous results for $\ell = 2$ to any prime ℓ and provides all data where the digraphs cannot be made undirected.
- (ii) We define superspecial isogeny digraphs $\mathfrak{G}_g(N, p)$ in full generality, and explore their properties. Our [Main Conjecture](#) describes the expansion properties of superspecial isogeny digraphs in dimension g .

- (iii) We computationally explore these superspecial digraphs $\mathfrak{G}_g(\ell, p)$ for a range of small primes for $g = 1$ and for $\ell = 2$ for $g = 2$. This gives us insight into the concrete expansion properties of these graphs, and isogeny walks in general. In particular, the impact of small cycles arising from endomorphisms of small degree is significant when $g = 1$. In higher dimensions, the resulting expansion properties differ significantly from previous work on undirected isogeny graphs.

We verify all propositions and theorems using **Magma**, and generate the superspecial isogeny digraph data for dimension $g = 1$ and $g = 2$ using **sage** [The24]. Our work can be used more generally to explore automorphisms in dimension 2 and (ℓ, ℓ) -level structures, and isogeny (di)graphs in dimension 1 and 2. The code and full data is available at:

<https://github.com/Krijn-math/good-graphs>

Table 1. The number of (ℓ, ℓ) -level Jacobians in function of their reduced automorphism group for primes $\ell > 3$. Clicking the group in the left column brings you to the associated theorem, which also covers $\ell = 2$ and $\ell = 3$.

RA(J)	number of (ℓ, ℓ) -level surfaces
1	$\ell^3 + \ell^2 + \ell + 1$
$\mathbb{Z}/2\mathbb{Z}$	$\frac{1}{2}(\ell^3 + 2\ell^2 + 3\ell + 2)$
$(\mathbb{Z}/2\mathbb{Z})^2$	$\frac{1}{4}(\ell^3 + 3\ell^2 + 6\ell + 5 + (\frac{-1}{\ell}))$
$\mathcal{S}_3$	$\frac{1}{6}(\ell^3 + 4\ell^2 + 9\ell + 8 + 2(\frac{\ell}{3}))$
$\mathcal{D}_{2 \times 6}$	$\frac{1}{12}(\ell^3 + 5\ell^2 + 14\ell + 19 + 3(\frac{-1}{\ell}) + 6(\frac{\ell}{3}))$
$\mathcal{S}_4$	$\frac{1}{24}(\ell^3 + 7\ell^2 + 24\ell + 41 + 9(\frac{-1}{\ell}) + 8(\frac{\ell}{3}) + 6(\frac{-2}{\ell}))$
$\mathbb{Z}/5\mathbb{Z}$	$\frac{1}{5}(\ell^3 + \ell^2 + \ell + 1 + \begin{cases} 4 & \text{if } \ell = 5 \\ 16 & \text{if } \ell \equiv 1 \pmod{5} \\ 0 & \text{if } \ell \not\equiv 1 \pmod{5} \end{cases})$

Notation and conventions. All fields are considered to be of positive characteristic $p > 5$, although some results may extend beyond that. We will denote ℓ for a (typically small) prime different from the field characteristic p . We will require k -th roots of unity both in $\mathbb{F}_p$ and $\mathbb{F}_\ell$ and will denote them by ζ_k and μ_k respectively. We will denote $\text{diag}(a_1, \dots, a_n)$ as the $n \times n$ diagonal matrix with entries $a_1, \dots, a_n$. Bases for (sub)spaces will be denoted by angled brackets $\langle \dots \rangle$ and will always be assumed to have the order fixed. We will denote E_0 and E_{1728} as the elliptic curves with respective j -invariants 0 and 1728. The n -th cyclotomic polynomial will be denoted by Φ_n and the dihedral group with $2n$ elements by $\mathcal{D}_{2 \times n}$. The notation $\left(\frac{p}{q}\right)$ is used for the Legendre symbol as per usual.

Table 2. The number of (ℓ, ℓ) -level elliptic products in function of their reduced automorphism group and prime ℓ . Clicking the group in the left column brings you to the associated theorem, which also covers $\ell = 2$ and $\ell = 3$.

$\text{RA}(E \times E')$	number of (ℓ, ℓ) -level surfaces
$\mathbb{Z}/2\mathbb{Z}$	$\frac{1}{2}(\ell^3 + 2\ell^2 + 3\ell + 2)$
$(\mathbb{Z}/2\mathbb{Z})^2$	$\frac{1}{4}(\ell^3 + 3\ell^2 + 6\ell + 5 + (\frac{-1}{\ell}))$
$\mathbb{Z}/6\mathbb{Z}$	$\frac{1}{6}(\ell^3 + 2\ell^2 + 7\ell + 6 + (4\ell + 4)(\frac{\ell}{3}))$
$\mathbb{Z}/4\mathbb{Z}$	$\frac{1}{4}(\ell^3 + 2\ell^2 + 5\ell + 4 + (2\ell + 2)(\frac{-1}{\ell}))$
$\mathbb{Z}/12\mathbb{Z}$	$\frac{1}{12}(\ell^3 + 2\ell^2 + 9\ell + 12 + 2(\ell + 3)(\frac{-1}{\ell}) + 4(\ell + 2)(\frac{\ell}{3}) + 4(\frac{3}{\ell}))$
$(\mathbb{Z}/2\mathbb{Z})^2 \times \mathbb{Z}/4\mathbb{Z}$	$\frac{1}{16}(\ell^3 + 4\ell^2 + 15\ell + 24 + (4\ell + 12)(\frac{-1}{\ell}) + 4(\frac{-2}{\ell}))$
$\mathbb{Z}/6\mathbb{Z} \times \mathcal{S}_3$	$\frac{1}{36}(\ell^3 + 5\ell^2 + 24\ell + 59 + (8\ell + 38)(\frac{\ell}{3}) + 9(\frac{-1}{\ell}))$

2 Preliminaries

We provide some of the necessary background regarding abelian varieties and expander (di)graphs. Remark that a large part of this paper specifically focuses on abelian surfaces, i.e. abelian varieties of dimension two, but some parts require the more general framework so we formulate them as such.

2.1 Abelian varieties

We work with abelian varieties over finite fields. A *polarization* of an abelian variety is an isogeny $\lambda : A \rightarrow \hat{A}$ between A and its dual $\hat{A}$, and such a polarization is *principal* if $\deg \lambda = 1$, i.e., we have an isomorphism $A \xrightarrow{\sim} \hat{A}$. An elliptic curve is naturally equipped with a principal polarization, and in general, the Jacobian of a curve of genus g is a principally polarized abelian variety of dimension g . Products of principally polarized abelian varieties $A \times B$ can be equipped with the *product polarization* $\lambda_{A \times B} = \lambda_A \times \lambda_B$. If not stated otherwise, we assume the product polarization on products of elliptic curves. A two-dimensional principally polarized abelian variety over $\overline{\mathbb{F}}_p$ is either (isomorphic to) the product of two elliptic curves, or the Jacobian of a hyperelliptic curve of genus 2. A *divisor* on an abelian variety is a formal finite integer sum of irreducible codimension-one subspaces. The divisors algebraically equivalent to zero (modulo linear equivalences) form the group $\text{Pic}^0(A)$ of an abelian variety. Concretely, for an abelian surface A constructed as the Jacobian of a genus-2 curve C , this means we can represent every element of A uniquely as the sum of at most two points of C .

Superspecial abelian varieties. In higher dimensions, supersingularity of elliptic curves generalizes to several distinct notions. Following [CDS20], we focus on superspecial abelian varieties. We say that an abelian variety A is *superspecial* if A is isomorphic to a product of supersingular curves, as an unpolarized variety. We say that a curve is superspecial when its accompanying Jacobian is superspecial.

A one-dimensional principally polarized abelian variety A over $\overline{\mathbb{F}}_p$ is an elliptic curve, and so, superspeciality reduces to supersingularity. All superspecial abelian varieties can be defined over $\mathbb{F}_{p^2}$.

Weil pairing. In general, the *Weil pairing of level n* on an abelian variety A is an alternating bilinear pairing $e_n : A[n] \times \hat{A}[n] \rightarrow \mu_n$. Given a (principal) polarization $\lambda : A \rightarrow \hat{A}$, we may define e_n^λ as the Weil pairing induced on $A[n] \times A[n]$. On (products of) elliptic curves or Jacobians of hyperelliptic curves, we assume the principal polarization and drop the superscript λ , to obtain the Weil pairing of level n as

$$e_n : A[n] \times A[n] \rightarrow \mu_n.$$

A basis $\langle P_1, \dots, P_g, Q_1, \dots, Q_g \rangle$ of $A[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$ is called *symplectic* iff

$$e_n(P_i, P_j) = e_n(Q_i, Q_j) = 1, \quad e_n(P_i, Q_j) = \mu_n^{\delta_{ij}},$$

where δ_{ij} is the Kronecker delta symbol. For an abelian surface specifically, the basis $\langle P_1, P_2, Q_1, Q_2 \rangle$ is symplectic iff the Weil pairing matrix in block form is

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell),$$

where entries need to be considered as $\log_n(e_n(\cdot, \cdot))$ and where $\mathrm{Sp}_4(\mathbb{F}_\ell)$ denotes the symplectic group of order four over $\mathbb{F}_\ell$.

Maximal isotropic subgroups. A subgroup $G \subseteq A[n]$ of a principally polarized abelian variety A is called *isotropic* with respect to the Weil pairing e_n^λ when $e_n^\lambda(P, Q) = 1$ for all $P, Q \in G$. If there is no larger isotropic subgroup $G' \subseteq A[n]$ such that $G \subsetneq G'$, then we say that G is a *maximal isotropic subgroup*. In this case, G is of rank g and has the structure $G \cong (\mathbb{Z}/n\mathbb{Z})^g$ as an abelian group. We refer to a set of points $P_1, \dots, P_g \in G$ as a *basis* of G whenever $G = \langle P_1, \dots, P_g \rangle$. An $(\ell, \dots, \ell)$ -*level structure* is a pair (A, G) , where G is a maximal isotropic $(\ell, \dots, \ell)$ -subgroup of A . For the rest of this work, we consider (A, G) *up to isomorphism*, which includes automorphisms.

Automorphisms. An *automorphism* α of a variety X/k is an isomorphism of X as a k -scheme. An automorphism α_* of an abelian variety A/k is a variety automorphism of A/k which is also a group homomorphism. The automorphisms of an abelian variety A form a group, which is denoted as $\mathrm{Aut}(A)$. An automorphism α of a curve C induces a natural automorphism α_* on the Jacobian of C by acting on the points of the divisor class. For (products of) elliptic curves, we will often write α instead of α_* due to the natural induced group law. Every abelian variety comes with the order-2 inversion automorphism $[-1]$ which maps D to $-D$. For this reason, we will typically work with the *reduced automorphism*

group of A , defined by $\text{RA}(A) = \text{Aut}(A)/[-1]$. We refer to [Appendix A](#) for a classification of the possible nontrivial automorphisms of p.p. abelian surfaces over finite fields. Automorphisms on abelian varieties preserve the Weil pairing; in particular they map maximal isotropic subgroups to maximal isotropic subgroups. An automorphism α_* acting on the ℓ -torsion $A[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^{2g}$ can be represented as a matrix $M \in \text{GL}_{2g}(\mathbb{Z}/\ell\mathbb{Z})$ by its action on a basis of $A[\ell]$.

Isogenies. We focus on separable $(\ell, \dots, \ell)$ -isogenies between principally polarized abelian varieties. Similar to dimension one, such isogenies are defined up to isomorphism by their kernel $G = \ker \varphi$. In this case, G must be a maximal isotropic subgroup of $A[\ell]$ isomorphic to $(\mathbb{Z}/\ell\mathbb{Z})^g$. By [\[CD22, Lemma 2\]](#), there are $\ell^{g(g+1)/2} \prod_{i=1}^g (1 + \frac{1}{\ell^i})$ maximal isotropic subgroups.

Given two $(\ell, \dots, \ell)$ -isogenies $\varphi_1 : A \rightarrow B$ and $\varphi_2 : B \rightarrow C$, we say that φ_2 is a *good extension* of φ_1 when $\ker \varphi_2 \cap \ker \hat{\varphi}_1 = 0$, following [\[CDS20\]](#). This implies that the composition $\varphi_2 \circ \varphi_1$ is an $(\ell^2, \dots, \ell^2)$ -isogeny. Two cases are important for this work:

- (i) In dimension one, any rank-1 subgroup $G \subseteq E[\ell]$ isomorphic to $\mathbb{Z}/\ell\mathbb{Z}$ is trivially maximal isotropic, as the Weil pairing will always be trivial on a rank-1 subgroup. When ℓ is prime, there are ℓ good extensions, as only the dual is eliminated. When ℓ is composite, good extensions can be viewed as those isogeny decompositions without backtracking.
- (ii) In dimension two, we mostly focus on $\ell = 2$. When $\ell = 2$, we have fifteen $(2, 2)$ -isogenies given by the fifteen maximal isotropic subgroups of $A[2]$. For a given $(2, 2)$ -isogeny $A \rightarrow B$, only eight out of the fifteen maximal isotropic subgroups of $B[2]$ define a good extension $B \rightarrow C$.

We distinguish four types of isogenies between abelian varieties. The most common is a *generic* isogeny $A \rightarrow B$ between simple abelian varieties A and B . A *splitting* isogeny is an isogeny $A \rightarrow B \times C$, and conversely, a *gluing* isogeny is an isogeny $A \times B \rightarrow C$. Lastly, a *diagonal* isogeny $A \times B \rightarrow C \times D$ is *de facto* two generic isogenies $A \rightarrow C$ and $B \rightarrow D$ evaluated simultaneously.

For computations of isogenies in dimension one, Vélú's formulas [\[Vél71\]](#) provide the necessary tools to compute an isogeny given a generator for its kernel. In dimension two, the computation of $(2, 2)$ -isogenies was given by Richelot [\[Ric37\]](#); for a more contemporary source, we refer to [\[Smi05\]](#).

2.2 Expander (di)graphs

Graphs. A graph $G = (V, E)$ consists of a set of vertices V and a set of edges E , which are unordered pairs of vertices. The set E is sometimes replaced by a multiset, which allows us to define multiple edges between the same two vertices. For our exposition, we will always allow this, but will only draw specific attention to it when necessary. A graph is called *k-regular* if for each vertex v there are k edges in E which have v as vertex.

A *walk* in a graph is a sequence of edges, where the “ending” vertex of an edge is the “starting” vertex of the subsequent edge. A graph is called *connected*

if there exists a walk between any two vertices. The *adjacency matrix* of a graph $G = (V, E)$ is an $n \times n$ matrix M , with $n = \#V$, where entry M_{ij} is the number of edges between vertex i and j , where we have chosen a fixed ordering for the vertices V . The adjacency matrix for a graph is symmetric since $M_{ij} = M_{ji}$.

Expansion properties. For cryptographic purposes, we are mostly interested in graphs where random walks are truly random in some sense. This type of behavior is labeled as *expander properties*. There are several notions of these, such as edge expansion, vertex expansion, and spectral expansion. Most types of expansion are related or can be reduced to one another and we will focus on spectral expansion since it is easiest to compute.

Let $G = (V, E)$ be a k -regular connected graph with $n = \#V$ and M its adjacency matrix. One can show that the eigenvalues λ_i of M satisfy $k = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -k$. The value $\lambda_1 - \lambda_2$ is called the spectral gap M . We call G a *Ramanujan graph* if

$$\lambda_2 \leq 2\sqrt{k-1}.$$

One may also define the Ramanujan property as a limit property for a family of graphs, which seems more apt for cryptographic purposes since we cannot compute eigenvalues of graphs with an exponential amount of nodes. We then say that a family of graphs $G_1, G_2, G_3, \dots$ has the Ramanujan property if for $i \rightarrow \infty$ it holds that there exists $\epsilon > 0$ such that for λ_2 of G_i we have $\lambda_2 \leq 2\sqrt{k-1} - \epsilon$. The value λ_2 also determines the rate of expansion.

Definition 1. A k -regular connected graph G has *spectral expansion* $\gamma(G)$ if

$$\lambda(G) = |\lambda_2/k| \leq 1 - \gamma(G),$$

where λ_2 is the eigenvalue of the adjacency matrix of G with second-largest absolute value.

Note that larger values of $\gamma(G)$ (or smaller values of $\lambda(G)$) correspond to better expansion properties.

Digraphs. A *directed graph or digraph* $G = (V, E)$ consists of a set of vertices V and a set of edges E , which are ordered pairs of vertices, as opposed to the unordered pairs of vertices for a graph. The *out-degree*, resp. *in-degree*, of a vertex $v \in V$ is the number of edges $(v_1, v_2) \in E$ such that $v = v_1$, resp. $v = v_2$. We will call a digraph k -regular if the out-degree for each vertex is k .

Most of the properties of graphs translate to digraphs, although subtleties do appear. The adjacency matrix of a digraph for example is no longer symmetric, hence its eigenvalues are not necessarily real. For a k -regular connected digraph, however, we still have that the largest eigenvalue λ_1 of its adjacency matrix equals k , which follows from the fact that the sum of the entries of each row is k . Given that the second largest eigenvalue λ_2 can now be complex, we say that a k -regular digraph G has spectral expansion $\gamma(G)$ if $\lambda(G) = |\lambda_2/k| \leq 1 - \gamma(G)$, where we now take the modulus instead of the absolute value. Similarly, we

can define a Ramanujan digraph as a k -regular connected digraph such that $|\lambda_2| \leq 2\sqrt{k-1}$.

The spectral expansion $\gamma(G)$ can be interpreted in several ways. The rapid mixing property essentially says that we can approach the uniform distribution using random walks as good as possible based on $\gamma(G)$.

Property 1. Let $G = (V, E)$ be a k -regular connected (di)graph with spectral expansion $\gamma(G)$ and $\#V = N$. Let $\pi^{(t)}$ be the distribution of the nodes of G after a random walk of length t , and let U_N be the uniform distribution on N elements. Then for every $\epsilon > 0$ the statistical distance $\frac{1}{2} \|\pi^{(t)} - U_N\|_1$ between $\pi^{(t)}$ and U_N is less than or equal to ϵ for

$$t \geq \frac{\log(\sqrt{N}/\epsilon)}{\log(1/\lambda(G))}.$$

Alternatively, we can also use the spectral expansion $\gamma(G)$ to express quasi-randomness, edge expansion, vertex expansion, etc. For all of these and more information about expander graphs and their properties in general, with a focus on directed graphs specifically, we refer to [DM, Vad].

3 Automorphisms acting on (ℓ, ℓ) -level surfaces

For a generic p.p. abelian surface A , it holds that the only nontrivial automorphism is $[-1]$, leading to $\text{RA}(A) = 1$. Since $[-1]$ maps any maximal isotropic subgroup of A to itself, an abelian surface with trivial $\text{RA}(A)$ gives rise to $\ell^3 + \ell^2 + \ell + 1$ distinct (ℓ, ℓ) -level surfaces.

In this section, we catalog the action of nontrivial automorphisms on (ℓ, ℓ) -level surfaces (A, G) . We discuss abelian surfaces A with *nontrivial* reduced automorphism group $\text{RA}(A)$, making a distinction between Jacobians of genus-2 curves (Section 3.2), and products of elliptic curves (Section 3.3).

Albeit that $[-1]$ acts trivially in $\text{RA}(A)$, we will sometimes use $[-1]$ in our proofs, in particular when there are other automorphisms α_* such that $\alpha_*^k = [-1]$ for some $k > 1$. Nonetheless, we will use the reduced automorphism group to make the comparison with [FS21] easier.

Throughout the proofs, we will also make use of Burnside's lemma, which we postulate here without proof. Recall that for an action of a group G on a set X we denote X/G as the set of orbits, and for every $g \in G$ the notation X^g corresponds to the set of points of X which are fixed under g .

Lemma 1. (*Burnside's lemma*) *Let G be a finite group acting on a finite set X . Then*

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

3.1 General action on symplectic spaces

Throughout the discussion, we will use the following lemma often without explicitly mentioning it.

Lemma 2. *Let γ_* be an automorphism of an n -dimensional abelian variety A , which acts as a linear map on $A[\ell]$ given by a matrix Γ for some basis $\mathcal{B}$ generating $A[\ell]$. If $\Gamma \in \mathrm{Sp}_{2n}(\mathbb{F}_\ell)$, then $\mathcal{B}$ is a symplectic basis.*

Proof. Choose a symplectic basis $\mathcal{B}'$ of $A[\ell]$. The automorphism γ_* acts as a linear map Γ' on $\mathcal{B}'$ and since automorphisms are pairing-preserving, we have that $\Gamma' \in \mathrm{Sp}_{2n}(\mathbb{F}_\ell)$. Thus Γ' and Γ are symplectically similar by a symplectic change of basis, which turns $\mathcal{B}'$ into a symplectic basis $\mathcal{B}$. $\square$

We will assume that all bases corresponding to an automorphism given in symplectic matrix form are symplectic unless specifically mentioned otherwise. We will also need the following lemmata.

Lemma 3. *Let V be a finite-dimensional K -vector space and M a K -linear map with eigenvalues $\lambda_1, \dots, \lambda_n$ and eigenspace decomposition $V = V_{\lambda_1} \oplus \dots \oplus V_{\lambda_n}$. Then a subspace $W \subseteq V$ is M -invariant iff*

$$W = (W \cap V_{\lambda_1}) \oplus \dots \oplus (W \cap V_{\lambda_n}).$$

Proof. Assume W is M -invariant and write $w = v_1 + \dots + v_n \in W$ with $v_i \in V_{\lambda_i}$. It suffices to show that each $v_i \in W$. Define the polynomial $p_i(x) = \prod_{j \neq i} (x - \lambda_j)$, then $p_i(\lambda_j) = 0$ when $i \neq j$ such that

$$p_i(M)w = p_i(M)v_1 + \dots + p_i(M)v_n = p_i(M)v_i = p_i(\lambda_i)v_i.$$

Since W is M -invariant, it is also $p_i(M)$ -invariant thus $p_i(\lambda_i)v_i \in W$ as wanted. The other direction of the proof is immediate. $\square$

Lemma 4. *Let $V = V_1 \oplus V_2$ be a finite-dimensional K -vector space with $\dim V_1 = \dim V_2 = n$ and let $M \in K^{2n \times 2n}$ be a K -linear map given by a square block matrix*

$$\begin{pmatrix} M_1 & 0 \\ 0 & M_2 \end{pmatrix}$$

such that $M_i V_i = V_i$. If V_1, V_2 are irreducible $K[M]$ -modules and M_1 and M_2 have the same characteristic polynomial, then the only n -dimensional M -invariant subspaces of V are V_1, V_2 and

$$\{(v_1, Lv_1) \mid v_1 \in V_1, L : V_1 \xrightarrow{\sim} V_2\},$$

for some isomorphism $L : V_1 \rightarrow V_2$. Furthermore, if M_1 and M_2 have different characteristic polynomials, then the only n -dimensional M -invariant subspaces of V are V_1 and V_2 .

Proof. Let $W \subseteq V$ be an n -dimensional M -invariant subspace and write $W_i = W \cap V_i$. Since W is M -invariant, the W_i are M_i -invariant and by irreducibility of V_i we have $\dim W_i \in \{0, n\}$. For $(\dim W_1, \dim W_2)$ equal to $(n, 0), (0, n), (n, n)$ we find that W equals V_1, V_2, V respectively, for which we need to discard the latter because its dimension is $2n$. So assume $\dim W_1 = \dim W_2 = 0$.

Since $W \cap V_2 = 0$, the projection $\pi_1 : W \rightarrow V_1$ is injective and by dimension argument it must be an isomorphism. The same holds for $\pi_2 : W \rightarrow V_2$. For any $v_1 \in V_1$ there is thus a unique $w \in W$ such that $\pi_1(w) = v_1$; define $L : V_1 \rightarrow V_2, v_1 \mapsto \pi_2(v_1)$. Then any w can be written as $(v_1, Lv_1) \in V_1 \oplus V_2$ and since π_2 is an isomorphism, so is L .

Now assume M_1 and M_2 have different characteristic polynomials; in particular they are not similar. Since W is M -invariant, we have that $(M_1 v_1, M_2 L v_1)$ must also be of the form $(v'_1, L v'_1)$ for some $v'_1 \in V_1$. Thus $M_2 L v_1 = L v'_1 = L M_1 v_1$ for all $v_1 \in V_1$ and consequently $M_2 L = L M_1$ as linear maps. Since M_1 and M_2 are not similar, no such L can exist. $\square$

3.2 Reduced automorphisms on Jacobians

Here, we go over the seven possible nontrivial reduced automorphism groups for Jacobians J , and discuss the action of these automorphisms on the (ℓ, ℓ) -subgroups of $J[\ell]$. For more details about the automorphisms, see [Appendix A](#) or [\[FS21\]](#).

RA(J) = $\mathbb{Z}/2\mathbb{Z}$ This case occurs if the Jacobian is $(2, 2)$ -split; i.e. there exists a $(2, 2)$ -isogeny $\Psi : J \rightarrow E \times E'$. Up to isomorphism, the curve equation giving rise to J can always be written in the form

$$C : y^2 = (x^2 - 1)(x^2 - s^2)(x^2 - t^2)$$

for suitable parameters s, t ; more specifically, $s \neq \pm t$ and $s, t \notin \{0, \pm 1\}$. The automorphism corresponding to $\mathbb{Z}/2\mathbb{Z}$ is given by $\sigma : x \mapsto -x$. We have the following characterization of the action of σ_* on (ℓ, ℓ) -level surfaces.

Proposition 1. *Let $C : y^2 = (x^2 - 1)(x^2 - s^2)(x^2 - t^2)$ be a genus-2 curve over a field of characteristic $p > 5$, with $\sigma : x \mapsto -x$ an automorphism of order 2, and $\ell \neq p$ a prime. Then the induced automorphism σ_* acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2
$\ell = 2$	7	8
$\ell \neq 2$	$(\ell + 1)^2$	$\ell^3 - \ell$

Proof. (i) For $\ell = 2$, see [\[FS21, Section 4.5\]](#).

- (ii) For $\ell \neq 2$, we know that $\sigma_*^2 = \text{id}$ so its minimal polynomial is $x^2 - 1$. The action on the ℓ -torsion is thus diagonalizable with eigenvalues ± 1 . Since the determinant of a symplectic matrix is always one, both must occur with multiplicity two such that σ_* can be given as $\text{diag}(1, -1, 1, -1) \in \text{Sp}_4(\mathbb{F}_\ell)$ for some symplectic basis. By [Lemma 3](#), the σ_* -invariant subspaces are thus built by choosing a line in the eigenspace corresponding to the eigenvalue 1 and a line in the eigenspace corresponding to the eigenvalue -1 , for which there are $(\ell + 1)^2$ options, all of which are maximal isotropic. $\square$

This leads to the following result.

Theorem 1. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with J its Jacobian, $\text{RA}(J) = \mathbb{Z}/2\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 11 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{2}(\ell^3 + 2\ell^2 + 3\ell + 2).$$

Proof. This follows immediately from [Proposition 1](#). $\square$

RA(J) = $(\mathbb{Z}/2\mathbb{Z})^2$ This is a specialization of the case for which $\text{RA}(J) = \mathbb{Z}/2\mathbb{Z}$, where it can be shown that any such J must stem from a curve given by the equation

$$C : y^2 = (x^2 - 1)(x^2 - u^2)(x^2 - 1/u^2),$$

for a suitable parameter $u \notin \{0, \pm 1, \pm i\}$. The automorphism corresponding to the additional component $\mathbb{Z}/2\mathbb{Z}$ is then given by $\tau : x \mapsto 1/x$. Remark that τ has order two in the reduced automorphism group only! Indeed, as a nonreduced automorphism on C , it induces τ_* on J for which it holds that $\tau_*^2 = [-1]$. We have the following characterization of the action of τ_* on (ℓ, ℓ) -level surfaces.

Proposition 2. *Let $C : y^2 = (x^2 - 1)(x^2 - u^2)(x^2 - 1/u^2)$ be a genus-2 curve over a field of characteristic $p > 5$, with $\tau : x \mapsto 1/x$ an automorphism of (reduced) order 2, and $\ell \neq p$ a prime. Then the induced automorphism τ_* acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces	
	in orbits of size 1	in orbits of size 2
$\ell = 2$	3	12
$\ell \equiv 1 \pmod{4}$	$\ell + 3$	$\ell^3 + \ell^2 - 2$
$\ell \equiv 3 \pmod{4}$	$\ell + 1$	$\ell^3 + \ell^2$

Proof. Since as a nonreduced automorphism we have that $\tau_*^2 = [-1]$, its minimal polynomial is $\Phi_4 = x^2 + 1$.

- (i) For $\ell = 2$, see [FS21, Section 4.7].
- (ii) For $\ell \equiv 1 \pmod{4}$, it holds that $\Phi_4 = (x - i)(x + i) \pmod{\ell}$, hence τ_* is diagonalizable with eigenvalues $\pm i$.^{*} Let D, D' be divisors with the same eigenvalue $\epsilon \in \{\pm i\}$, we then find that

$$e_\ell(D, D') = e_\ell(\tau_*(D), \tau_*(D')) = e_\ell(\epsilon D, \epsilon D') = e_\ell(D, D')^{-1},$$

from which it follows that the pairings must be trivial since ℓ is odd. The non-degeneracy of the e_ℓ -pairing implies that both eigenvalues have eigenspaces (say V_i and V_{-i}) of dimension two, such that τ_* is given by $\text{diag}(i, -i, -i, i) \in \text{Sp}_4(\mathbb{F}_\ell)$. Using Lemma 3, we see that for every line in V_i there is exactly one line in V_{-i} making a maximal isotropic subgroup. Combining with the two maximal isotropic subgroups V_i and V_{-i} themselves, we thus have $\ell + 3$ τ_* -invariant maximal isotropic subgroups.

- (iii) For $\ell \equiv 3 \pmod{4}$, we have that Φ_4 is irreducible modulo ℓ . For a suitable choice of symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$, we thus have that τ_* is given by

$$\begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell).$$

By Lemma 4, there are $\ell + 1$ choices of graphs of isomorphisms $\langle D_1, D_2 \rangle \rightarrow \langle D'_1, D'_2 \rangle$ which are τ_* -invariant two-dimensional subspaces. These are lines corresponding to the subgroups $\langle aD_1 + bD'_1, aD_2 + bD'_2 \rangle$, where a and b are not simultaneously zero. Since the basis is symplectic, the only nontrivial pairings are $e_\ell(D_1, D'_1)$ and $e_\ell(D_2, D'_2)$, from which one can easily deduce that all of these τ_* -invariant subgroups are isotropic.

□

Using the action of σ_* from Proposition 1, we have the following theorem.

Theorem 2. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with J its Jacobian, $\text{RA}(J) = (\mathbb{Z}/2\mathbb{Z})^2$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 8 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{4} \left(\ell^3 + 3\ell^2 + 6\ell + 5 + \left(\frac{-1}{\ell} \right) \right).$$

Proof. Burnside's lemma says that the number of (ℓ, ℓ) -level surfaces is equal to

$$\frac{1}{|(\mathbb{Z}/2\mathbb{Z})^2|} \cdot (|X^{\text{id}}| + |X^{\sigma_*}| + |X^{\tau_*}| + |X^{\sigma_*\tau_*}|),$$

hence by Propositions 1 and 2, it suffices to check $X^{\sigma_*\tau_*}$. One can verify that the minimal polynomial of $\sigma_*\tau_*$ is $x^2 - 1$, so the orbits act exactly the same as σ_* from Proposition 1.

□

^{*}Remark that by i we mean i such that $i^2 \equiv -1 \pmod{\ell}$. Moral principles prevent us from using μ_4 .

$\mathbf{RA}(J) = \mathcal{S}_3$ This is a one-parameter family of curves C which are a specialization of the case for which the reduced automorphism group is $\mathbb{Z}/2\mathbb{Z}$. More specifically, define

$$C : y^2 = (x^2 - 1)(x^2 - s^2)(x^2 - t^2),$$

with

$$s = \frac{(v+1)(v-\zeta_3)}{(v-1)(v+\zeta_3)}, \quad t = \frac{(v+1)(v-\zeta_3^2)}{(v-1)(v+\zeta_3^2)},$$

where v is a free parameter and ζ_3 a cubic root of unity. The extra automorphism of order 3 is given by

$$\rho : x \mapsto \frac{(2\zeta_3 + 1)(v^2 - 1)x + 3(v+1)^2}{3(v-1)^2x + (2\zeta_3 + 1)(v^2 - 1)}.$$

We have the following characterization of the action of ρ_* on (ℓ, ℓ) -level surfaces.

Proposition 3. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with $\mathbf{RA}(C) = \mathcal{S}_3$, with ρ an automorphism of order 3 (given as above), and $\ell \neq p$ a prime. Then the induced automorphism ρ_* acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 3
$\ell = 3$	7	33
$\ell \equiv 1 \pmod{3}$	$\ell + 3$	$\ell^3 + \ell^2 - 2$
$\ell \equiv 2 \pmod{3}$	$\ell + 1$	$\ell^3 + \ell^2$

Proof. Since $\rho_*^3 = \text{id}$, its minimal polynomial is $\Phi_3 = x^2 + x + 1$.

- (i) For $\ell = 3$ we have that ρ_* is nilpotent of order three since $(x-1)^3 = x^3 - 1 \pmod{3}$ hence the only ρ_* -invariant subspaces are those in the kernel of $(\rho_* - 1)^k$ for some $k \in \{1, 2, 3\}$. The ρ -invariant points of C are

$$Q_{\pm, \pm} = \left(\pm \frac{v+1}{v-1}, \pm 8 \cdot \frac{v\sqrt{v}(v+1)^2}{(v-1)^3(v+\zeta_3)(v+\zeta_3^2)} \right).$$

One can verify that the ρ^2 -invariant points on C are also just $Q_{\pm, \pm}$, since the map ρ^2 is the same as ρ apart from the factors $(2\zeta_3 + 1)$ now having a minus sign in front. From these we can build exactly two one-dimensional ρ_* -invariant subspaces, say

$$H_+ = \langle Q_{+,+} - Q_{+,-} \rangle, \quad H_- = \langle Q_{-,+} - Q_{-,-} \rangle.$$

One can verify that these have trivial 3-Weil pairing, hence ρ_* is given by

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell)$$

for some symplectic basis. We immediately see that $\langle H_+, H_- \rangle$ as well as the upper left block and the bottom right block create three maximal isotropic ρ_* -invariant subspaces. Remark that the blocks are *not* irreducible, hence [Lemma 4](#) does not hold. However, following the proof of [Lemma 4](#), it is clear we only need to consider the 1-dimensional subspace case separately (which we have by means of H_+ and H_-), so the other subspaces are created by isomorphisms between the blocks. These correspond to lines so there are $\ell + 1$ of them and the isotropy condition follows from the symplectic basis.

- (ii) For $\ell \equiv 1 \pmod 3$, a primitive cubic root of unity $\mu_3 \in \mathbb{F}_\ell$ exists such that Φ_3 splits completely modulo ℓ and ρ_* is diagonalizable with eigenvalues μ_3 and μ_3^2 . We have that ρ_* must be given by $\mathrm{diag}(\mu_3, \mu_3, \mu_3^2, \mu_3^2) \in \mathrm{Sp}_4(\mathbb{F}_\ell)$ and we have $\ell + 3$ maximal isotropic subgroups which are ρ_* -invariant just as in [Proposition 2\(ii\)](#).
- (iii) For $\ell \equiv 2 \pmod 3$, Φ_3 is irreducible modulo ℓ and for a suitable choice of basis $\langle D_1, D_2, D'_1, D'_2 \rangle$ we can write that ρ_* acts as

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell).$$

Remark that this matrix is in “blocks” acting on $\langle D_1, D'_1 \rangle$ and $\langle D_2, D'_2 \rangle$. By [Lemma 4](#), the only possible ρ_* -invariant maximal isotropic subspaces are $\langle D_1, D'_1 \rangle$, $\langle D_2, D'_2 \rangle$ (which are not isotropic) and all graphs consisting of isomorphisms $\langle D_1, D'_1 \rangle \rightarrow \langle D_2, D'_2 \rangle$. There are $\ell + 1$ of those just as in [Proposition 2\(iii\)](#).

□

Since $\mathrm{RA}(J) = \langle \sigma_*, \rho_* \rangle$, we have the following result.

Theorem 3. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with J its Jacobian, $\mathrm{RA}(J) = \mathcal{S}_3$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 7 if $\ell = 2$, equal to 17 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{6} \left(\ell^3 + 4\ell^2 + 9\ell + 8 + 2 \left(\frac{\ell}{3} \right) \right).$$

Proof. Burnside’s lemma says that the number of (ℓ, ℓ) -level surfaces is equal to

$$\frac{1}{|\mathcal{S}_3|} \cdot \left(|X^{\mathrm{id}}| + |X^{\sigma_*}| + |X^{\rho_*}| + |X^{\rho_*^2}| + |X^{\rho_*\sigma_*}| + |X^{\rho_*^2\sigma_*}| \right).$$

By [Propositions 1 and 3](#) and it suffices to check $X^{\rho_*\sigma_*}$ and $X^{\rho_*^2\sigma_*}$. One can verify that the minimal polynomial of both $\rho_*\sigma_*$ and $\rho_*^2\sigma_*$ is $x^2 - 1$, so the orbits act exactly the same as σ_* from [Proposition 1](#). $\square$

RA(J) = $\mathcal{D}_{2 \times 6}$ This case only occurs if the Jacobian stems from the genus-2 curve defined by $C : y^2 = x^6 + 1$, and the automorphism is given by $\zeta^{(6)} : x \mapsto \zeta_6 x$, where ζ_6 is a primitive sixth root of unity in $\overline{\mathbb{F}}_p$. This is a special case of $\text{RA}(C) = (\mathbb{Z}/2\mathbb{Z})^2$, where $(\zeta^{(6)})^3 = \sigma : x \mapsto -x$, such that $\text{RA}(C) = \langle \tau, \zeta^{(6)} \rangle$. The curve is superspecial iff $p \equiv 5 \pmod{6}$, see [\[IKO86, Proposition 1.11\]](#). We have the following characterization of the action of $\zeta_*^{(6)}$ on (ℓ, ℓ) -level surfaces.

Proposition 4. *Let $C : y^2 = x^6 + 1$ be a genus-2 curve over a field of characteristic $p > 5$, with $\zeta^{(6)} : x \mapsto \zeta_6 x$ an automorphism of order 6, and $\ell \neq p$ a prime. Then the induced automorphism $\zeta_*^{(6)}$ acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2	(ℓ, ℓ) -level surfaces in orbits of size 3	(ℓ, ℓ) -level surfaces in orbits of size 6
$\ell = 2$	1	2	6	6
$\ell = 3$	1	6	15	18
$\ell \equiv 1 \pmod{3}$	4	$\ell - 1$	$(\ell + 1)^2 - 4$	$\ell^3 - 2\ell + 1$
$\ell \equiv 2 \pmod{3}$	0	$\ell + 1$	$(\ell + 1)^2$	$\ell^3 - 2\ell - 1$

Proof. Since $\zeta_*^{(6)} = \text{id}$, its minimal polynomial must divide $x^6 + 1$. One can verify that it equals $x^4 + x^2 + 1 = (x^2 - x + 1)(x^2 + x + 1) = \Phi_6\Phi_3$.

- (i) See [\[FS21, Section 4.11\]](#).
- (ii) For $\ell = 3$ the minimal polynomial reduces to $(x - 1)^2(x + 1)^2 \pmod{3}$. Given that this concerns one specific ℓ for one specific curve (up to isomorphism), the count is a direct computation. For the sake of completeness, the symplectic matrices of $\zeta_*^{(6)}$, $(\zeta_*^{(6)})^2$ and $(\zeta_*^{(6)})^3$ are given by respectively

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

- Remark that the last one equals the one of [Proposition 1](#) since $(\zeta_*^{(6)})^3 = \sigma_*$.
- (iii) For $\ell \equiv 1 \pmod{3}$, a primitive cubic root of unity $\mu_3 \in \mathbb{F}_\ell$ exists such that the minimal polynomial splits completely modulo ℓ , whence $\zeta_*^{(6)}$ acts as $\text{diag}(\mu_3, -\mu_3, \mu_3^2, -\mu_3^2) \in \text{Sp}_4(\mathbb{F}_\ell)$ on some basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. By [Lemma 3](#), the $\zeta_*^{(6)}$ -invariant maximal isotropic subspaces are hence direct

sums of two one-dimensional eigenspaces. We thus have four $\zeta_*^{(6)}$ -invariant maximal isotropic subspaces:

$$\langle D_1, D_2 \rangle, \langle D_1, D'_2 \rangle, \langle D'_1, D_2 \rangle, \langle D'_1, D'_2 \rangle.$$

Squaring this matrix results in $\text{diag}(\mu_3^2, \mu_3^2, \mu_3, \mu_3) \in \text{Sp}_4(\mathbb{F}_\ell)$. By checking the e_ℓ -pairing, one can readily verify that any two-dimensional subspace consisting of a line in the eigenspace with eigenvalue μ_3^2 has exactly one line in the eigenspace with eigenvalue μ_3 which creates a maximal isotropic subgroup. There are thus $\ell - 1$ $(\zeta_*^{(6)})^2$ -invariant subspaces (since we already counted $\langle D_1, D'_2 \rangle$ and $\langle D_2, D'_1 \rangle$). We also know that $(\zeta_*^{(6)})^3 = \sigma_*$, which has $(\ell + 1)^2$ invariant maximal isotropic subspaces. Remark that we need to subtract the $\zeta_*^{(6)}$ -invariant ones from this count, but not the $(\zeta_*^{(6)})^2$ -invariant ones since these do not overlap.

- (iv) For $\ell \equiv 2 \pmod 3$, the minimal polynomial is irreducible modulo ℓ and for a suitable choice of basis $\langle D_1, D_2, D'_1, D'_2 \rangle$ we can write that $\zeta_*^{(6)}$ acts as

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell),$$

which decomposes as a direct summand as it acts in “blocks” on $\langle D_1, D'_1 \rangle$ and $\langle D_2, D'_2 \rangle$. Since they are not similar, it follows from [Lemma 4](#) that the only $\zeta_*^{(6)}$ -invariant subspaces are $\langle D_1, D'_1 \rangle$ and $\langle D_2, D'_2 \rangle$ but those are not isotropic. The action of $(\zeta_*^{(6)})^2$ is then given by

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell),$$

where the “blocks” are now similar, hence by [Lemma 4](#) the $\ell + 1$ isomorphisms between them provide the $(\zeta_*^{(6)})^2$ -invariant maximal isotropic subgroups. The case $(\zeta_*^{(6)})^3 = \sigma_*$ is similar as before.

□

Combining this proposition with [Proposition 2](#) we get the following result.

Theorem 4. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with J its Jacobian, $\text{RA}(J) = \mathcal{D}_{2 \times 6}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 5 if $\ell = 2$, equal to 11 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{12} \left(\ell^3 + 5\ell^2 + 14\ell + 19 + 3 \left(\frac{-1}{\ell} \right) + 6 \left(\frac{\ell}{3} \right) \right).$$

Proof. Burnside's lemma says that the number of (ℓ, ℓ) -level surfaces is equal to

$$\frac{1}{|\mathcal{D}_{2 \times 6}|} \cdot \left(|X^{\text{id}}| + \sum_{k=1}^5 |X^{(\zeta_*^{(6)})^k}| + |X^{\tau_*}| + \sum_{i=1}^5 |X^{\tau_* (\zeta_*^{(6)})^i}| \right).$$

By [Propositions 2](#) and [4](#) we need only verify the $X^{\tau_* (\zeta_*^{(6)})^k}$ for $1 \leq k \leq 5$. For $k = 3$ we have that $\tau_* (\zeta_*^{(6)})^3_* = \tau_* \sigma_*$, which has the same number of fixed points as σ_* as we have shown in the proof of [Theorem 2](#). One can verify that the minimal polynomial of $\tau_* \zeta_*^{(6)}$ and $\tau_* (\zeta_*^{(6)})^5$ is $x^2 - 1$, so the orbits act exactly the same as σ_* from [Proposition 1](#). Similarly, the minimal polynomial of $\tau_* (\zeta_*^{(6)})^2$ and $\tau_* (\zeta_*^{(6)})^4$ is $x^2 + 1$, so the orbits act exactly the same as τ_* from [Proposition 2](#). $\square$

RA(J) = $\mathcal{S}_4$ This is a special case of $\text{RA}(C) = \mathcal{S}_3$ where we need to choose

$$v = \frac{1}{\sqrt{2}}(\zeta_{12}^2 + \zeta_{12} + 1).$$

Remark that there exists a $(2, 2)$ -isogeny from the Jacobian of C to E^2 , with $j(E) = 8000$. The extra automorphism to generate $\mathcal{S}_4$ is given by

$$\omega : x \mapsto \frac{x - (\sqrt{2} + 1)}{(\sqrt{2} - 1)x + 1}.$$

Since this is a specialization of a previous case, we have that $\text{RA}(C) = \mathcal{S}_4 = \langle \sigma, \rho, \omega \rangle = \langle \rho, \omega \rangle$, where ω has order four. The curve is superspecial iff $p \equiv 5, 7 \pmod{8}$, see [\[IKO86, Proposition 1.12\]](#). Note that as a nonreduced automorphism, ω has order eight![†] We have the following characterization of the action of ω_* on (ℓ, ℓ) -level surfaces.

Proposition 5. *Let C be a genus-2 curve over a field of characteristic $p > 5$, with ω an automorphism of order 8, and $\ell \neq p$ a prime. Then the induced automorphism ω_* acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2	(ℓ, ℓ) -level surfaces in orbits of size 4
$\ell = 2$	1	2	12
$\ell \equiv 1 \pmod{8}$	4	$\ell - 1$	$\ell^3 + \ell^2 - 2$
$\ell \equiv 3 \pmod{8}$	2	$\ell - 1$	$\ell^3 + \ell^2$
$\ell \equiv 5 \pmod{8}$	2	$\ell + 1$	$\ell^3 + \ell^2 - 2$
$\ell \equiv 7 \pmod{8}$	0	$\ell + 1$	$\ell^3 + \ell^2$

[†]The curve in question is isomorphic to $C : y^2 = x^5 - x$, with the new automorphism given by $(x, y) \mapsto (ix, \sqrt{i}y)$. In a vacuum, it would be easier to work with this defining equation, but we will use the other automorphisms up ahead as well so we stick with the choice of v . Remark that over a field of characteristic five, this curve actually has $\text{RA}(C) = \text{PGL}(2, 5) \cong \mathcal{S}_5$ [\[IKO86\]](#).

Proof. Since $\omega_*^4 = [-1]$, its minimal polynomial is $\Phi_8 = x^4 + 1$.

- (i) For $\ell = 2$, see [FS21, Section 4.14].
- (ii) For $\ell \equiv 1 \pmod 8$ we have that Φ_8 splits completely modulo ℓ , hence the action of ω_* is given by $\text{diag}(\mu_8, \mu_8^3, \mu_8^7, \mu_8^5) \in \text{Sp}_4(\mathbb{F}_\ell)$ for some symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. Combining Lemma 2 and Lemma 3, the ω_* -invariant maximal isotropic subgroups are thus given by

$$\langle D_1, D_2 \rangle, \langle D_1, D'_1 \rangle, \langle D'_1, D_2 \rangle, \langle D'_1, D'_2 \rangle.$$

For ω_*^2 we have that the action is given by $\text{diag}(i, -i, -i, i) \in \text{Sp}_4(\mathbb{F}_\ell)$. Any line in the eigenspace corresponding to i pairs with exactly one line in the eigenspace corresponding to $-i$, as well as the two eigenspaces themselves. We already counted four of them as ω_* -invariant, so there are $\ell - 1$ remaining.

- (iii) For $\ell \equiv 3 \pmod 8$ we have that $\Phi_8 = (x^2 + \sqrt{-2}x - 1)(x^2 - \sqrt{-2}x - 1) \pmod \ell$. The actions of ω_* and ω_*^2 are thus given by the matrices

$$\begin{pmatrix} -\sqrt{-2} & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & \sqrt{-2} \end{pmatrix}, \quad \begin{pmatrix} -1 & -\sqrt{-2} & 0 & 0 \\ -\sqrt{-2} & 1 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{-2} \\ 0 & 0 & \sqrt{-2} & -1 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell)$$

for some basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. Since the blocks for ω_* are *not* similar but those for ω_*^2 are, by Lemma 4 we know that there are $\ell + 1$ ω_*^2 -invariant maximal isotropic subspaces; the blocks themselves give ω_* -invariant maximal isotropic subspaces $\langle D_1, D_2 \rangle, \langle D'_1, D'_2 \rangle$.

- (iv) For $\ell \equiv 5 \pmod 8$ we have that $\Phi_8 = (x^2 + i)(x^2 - i) \pmod \ell$ such that ω_* is given by

$$\begin{pmatrix} 0 & -i & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & 1 & 0 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell)$$

for some basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. Given how the blocks are not similar, we have that $\langle D_1, D_2 \rangle$ and $\langle D'_1, D'_2 \rangle$ are the only ω_* -invariant maximal isotropic subspaces by Lemma 4. For ω_*^2 we immediately see that the action is given by $\text{diag}(-i, -i, i, i) \in \text{Sp}_4(\mathbb{F}_\ell)$, and by Lemma 3 we can combine any line corresponding to the eigenspace with eigenvalue $-i$ with any line corresponding to the eigenspace with eigenvalue i , keeping in mind we need to discard the two that were already ω_* -invariant.

- (v) For $\ell \equiv 7 \pmod 8$ we have that $\Phi_8 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1) \pmod \ell$ such that ω_* and ω_*^2 are given by

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -\sqrt{2} & 0 \\ 0 & -1 & 0 & \sqrt{2} \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 & -\sqrt{2} & 0 \\ 0 & -1 & 0 & \sqrt{2} \\ \sqrt{2} & 0 & 1 & 0 \\ 0 & -\sqrt{2} & 0 & 1 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell).$$

The argument is the same as for $\ell \equiv 3 \pmod 8$, except now the blocks themselves are only ω_*^2 -invariant and not ω_* -invariant.

□

We have that $\text{RA}(J) = \langle \rho_*, \omega_* \rangle$, resulting in the following theorem.

Theorem 5. *Let C be a genus-2 curve over a field of characteristic $p > 5$ with J its Jacobian, $\text{RA}(J) = \mathcal{S}_4$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 4 if $\ell = 2$, equal to 9 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{24} \left(\ell^3 + 7\ell^2 + 24\ell + 41 + 9 \left(\frac{-1}{\ell} \right) + 8 \left(\frac{\ell}{3} \right) + 6 \left(\frac{-2}{\ell} \right) \right).$$

Proof. One can verify that the group $\mathcal{S}_4$ contains

- one identity element id ;
- six elements of order two (the transpositions) with minimal polynomial $x^2 - 1$ acting as σ_* from [Proposition 1](#);
- three elements of order two (two disjoint transpositions) with minimal polynomial $x^2 + 1$ acting as τ_* from [Proposition 2](#);
- eight elements of order three with minimal polynomial $x^2 + x + 1$ acting as ρ_* from [Proposition 3](#);
- six elements of order four with minimal polynomial $x^4 + 1$ acting as ω_* from [Proposition 5](#).

We thus have that the number of (ℓ, ℓ) -level surfaces is given by

$$\frac{1}{24} \cdot (|X^{\text{id}}| + 6|X^{\sigma_*}| + 3|X^{\tau_*}| + 8|X^{\rho_*}| + 6|X^{\omega_*}|).$$

□

RA(J) = $\mathbb{Z}/5\mathbb{Z}$ This case only occurs if the Jacobian stems from the genus-2 curve defined by $C : y^2 = x^5 - 1$, and the automorphism is given by $\zeta^{(5)} : x \mapsto \zeta_5 x$, where ζ_5 is a primitive fifth root of unity in $\overline{\mathbb{F}}_p$. The curve is superspecial iff $p \equiv 4 \pmod{5}$, see [\[IKO86, Proposition 1.13\]](#). We have the following characterization of the action of $\zeta_*^{(5)}$ on (ℓ, ℓ) -level surfaces.

Proposition 6. *Let $C : y^2 = x^5 - 1$ be a genus-2 curve over a field of characteristic $p > 5$, with $\zeta^{(5)} : x \mapsto \zeta_5 x$ an automorphism of order 5, and $\ell \neq p$ a prime. Then the induced automorphism $\zeta_*^{(5)}$ acts on the (ℓ, ℓ) -level surfaces of C as follows:*

	(ℓ, ℓ) -level surfaces	
	in orbits of size 1	in orbits of size 5
$\ell = 5$	1	155
$\ell \equiv 1 \pmod{5}$	4	$\ell^3 + \ell^2 + \ell - 3$
$\ell \not\equiv 1 \pmod{5}$	0	$\ell^3 + \ell^2 + \ell + 1$

Proof. Since $\zeta_*^{(5)} = \text{id}$, its minimal polynomial is $\Phi_5 = x^4 + x^3 + x^2 + x + 1$.

- (i) If $\ell = 5$ then $\Phi_5 = (x - 1)^4 \bmod 5$ and the only $\zeta_*^{(5)}$ -invariant subspaces are those in the kernel of $(\zeta_*^{(5)} - 1)^k$ for some $k \in \{1, 2, 3, 4\}$. The only points on the curve invariant under $\zeta^{(5)}$ are given by $Q = (0, i)$ and $Q' = (0, -i)$, giving rise to the $\zeta_*^{(5)}$ -invariant one-dimensional subspace

$$\langle [Q - \infty] \rangle = \{0, [Q - \infty], [2Q - 2\infty], [2Q' - 2\infty], [Q' - \infty]\} \subseteq J[5].$$

Any other divisor $[P + Q - 2\infty]$ cannot be invariant under $\zeta_*^{(5)}$ as this requires either $\zeta^{(5)}(P) = P$, as we had before, or $\zeta^{(5)}(P) = Q$ and vice versa, which implies $x_P = \zeta_5^2 x_P$ hence $x_P = 0$. As we only have a single eigenvector for $\lambda = 1$, we can thus assume that $\zeta_*^{(5)}$ is given by

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell).$$

By [Lemma 3](#), the only $\zeta_*^{(5)}$ -invariant two-dimensional subspace stems from $(\zeta_*^{(5)} - 1)^2$, which is maximal isotropic since $\zeta_*^{(5)}$ is symplectic.

- (ii) If $\ell \equiv 1 \bmod 5$, then a primitive fifth root of unity exists in $\mathbb{F}_\ell$, such that $\Phi_5 = \prod_{i=1}^4 (x - \mu_5^i)$ splits completely modulo ℓ . We can thus assume that $\zeta_*^{(5)}$ is given by $\text{diag}(\mu_5, \mu_5^2, \mu_5^4, \mu_5^3) \in \text{Sp}_4(\mathbb{F}_\ell)$ for some basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. Combining [Lemma 2](#) and [Lemma 3](#), the $\zeta_*^{(5)}$ -invariant maximal isotropic subgroups are thus given by

$$\langle D_1, D_2 \rangle, \langle D_1, D'_2 \rangle, \langle D'_1, D_2 \rangle, \langle D'_1, D'_2 \rangle.$$

- (iii) If $\ell \equiv 2, 3 \bmod 5$ then Φ_5 is irreducible modulo ℓ and no $\zeta_*^{(5)}$ -invariant two-dimensional subspace of $J[\ell]$ exists. If $\ell \equiv 4 \bmod 5$, a fifth root of unity exists modulo ℓ^2 , hence

$$\Phi_5 = (x^2 - (\mu_5 + \mu_5^4)x + 1) \cdot (x^2 - (\mu_5^2 + \mu_5^3)x + 1) \bmod \ell.$$

We can thus assume that $\zeta_*^{(5)}$ is given by the matrix

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & \mu_5 + \mu_5^4 & 0 \\ 0 & 1 & 0 & \mu_5^2 + \mu_5^3 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell).$$

for some symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$. Thus it decomposes as a direct summand of the two irreducible subspaces $\langle D_1, D'_1 \rangle$ and $\langle D_2, D'_2 \rangle$. Since the “blocks” are not similar, it follows from [Lemma 4](#) that these irreducible spaces are the only ones to consider, but they are not symplectic.

□

This immediately results in the following proposition.

Theorem 6. *Let C be a genus-2 curve over a field of characteristic $p > 5$, J its Jacobian, with $\text{RA}(J) = \mathbb{Z}/5\mathbb{Z}$. Then the number of (ℓ, ℓ) -level surfaces (J, G) is equal to 32 if $\ell = 5$ and*

- $\frac{1}{5}(\ell^3 + \ell^2 + \ell + 17)$ if $\ell \equiv 1 \pmod{5}$;
- $\frac{1}{5}(\ell^3 + \ell^2 + \ell + 1)$ if $\ell \not\equiv 1 \pmod{5}$.

Proof. This follows from [Proposition 6](#) and Burnside’s lemma. □

3.3 Reduced automorphisms on products of elliptic curves

Every product of elliptic curves has $\text{RA}(E \times E') \neq 1$, since $[1] \times [-1]$ is always nontrivial. There are seven possible (reduced) automorphism groups of elliptic curves; see [Appendix A](#) or [\[FS21\]](#). The six other options all stem from either both factors being isomorphic, which gives the automorphism $(P, Q) \mapsto (Q, P)$, or one of the factors having the automorphism $\iota : E_{1728} \rightarrow E_{1728}, (x, y) \mapsto (-x, iy)$ or $\zeta^{(3)} : E_0 \rightarrow E_0, (x, y) \mapsto (\zeta_3 x, y)$. The curves E_{1728} and E_0 are supersingular if $p \equiv 3 \pmod{4}$, respectively $p \equiv 2 \pmod{3}$, and can thus make part of a superspecial abelian surface. To not distract from the discussion, we refer to [Appendix B](#) for the full details since combining the automorphisms ι and $\zeta^{(3)}$ is long and technical rather than insightful.

3.4 Computational verification

Although the results in [Sections 3.2](#) and [3.3](#) and [Tables 1](#) and [2](#) are rigorously proven, we have simultaneously computationally verified these results for all possible reduced automorphism groups for $2 \leq \ell \leq 79$ for a range of primes p . This ensures that no minor mistakes or typos have been made in counting arguments. This section briefly describes this verification process, as the computational techniques may be applied more generally when working with (ℓ, ℓ) -subgroups and (ℓ, ℓ) -isogenies.

Assume a supersingular principally polarized abelian surface A over $\mathbb{F}_{p^2}$ with $A(\mathbb{F}_{p^2}) = A[p+1]$ and a prime $\ell \mid p+1$ such that $A[\ell]$ is rational. Furthermore, assume $\alpha_* \in \text{RA}(A)$ is a non-trivial automorphism. Following [\[KTW22\]](#), we may construct a symplectic basis $\mathcal{B} := \langle D_1, D_2, D'_1, D'_2 \rangle$ for $A[\ell]$. Using the echelon form ([Appendix C](#)), we may generate the set $\mathcal{G}$ of all valid maximally isotropic subgroups $G = \langle K_1, K_2 \rangle$ as linear combinations in $\mathcal{B}$, and let α_* act on both K_i to derive $\alpha_*(G)$. However, this procedure is time consuming, and verifying equality of $\alpha_*(G)$ and $\alpha_*(G')$ is non-trivial. Instead, we compute the matrix M_α from the action of α_* on $\mathcal{B}$ and let M_α directly act on the echelon matrix M_G associated to G . Verifying equality is now easy using the reduced echelon form of $M_\alpha \cdot M_G$.

This allows us to split $\mathcal{G}$ into its orbits under α_* , which allows us to verify the propositions in [Sections 3.2](#) and [3.3](#). Similarly, to verify the theorems, we split $\mathcal{G}$ into its orbits under the full group $\text{RA}(A)$ such that the number of orbits gives us the number of (A, G) up to isomorphism. The **Magma** code that verifies these results for all Jacobians and products with non-trivial reduced automorphism groups is available in our repository.

4 Expansion properties of isogeny digraphs

In this section, we describe the expansion properties of directed isogeny graphs. We first define the general superspecial isogeny *digraph* for any dimension g , before discussing the experimental data obtained on the eigenvalues of the adjacency matrix for dimension 1, for several small primes, and dimension 2, for $\ell = 2$. This allows us to state our [Main Conjecture](#) on the concrete expansion properties of superspecial isogeny digraphs, which determine the rate at which a random walk approaches the uniform distribution.

4.1 The Superspecial $(\ell)^g$ -isogeny Digraph

We define the superspecial $(\ell)^g$ -isogeny digraph in its full generality.

Definition 2. The *superspecial $(\ell)^g$ -isogeny digraph* $\mathfrak{G}_g(p, \ell) = (V, E)$ is defined as follows.

- The vertex set V consists of all isomorphism classes (A, H) , where A is a g -dimensional superspecial abelian variety over $\overline{\mathbb{F}}_p$ and $H \subseteq A[\ell]$ is a maximal isotropic subgroup.
- The edge set E consists of all ordered pairs $((A_1, H_1), (A_2, H_2))$ for which there exists an isogeny $\Psi_1 : A_1 \rightarrow A_2$ with $\ker \Psi_1 \cap H_1 = 0$ and $\Psi_1(H_1) = H_2$.

For $g = 1$, Charles, Goren and Lauter [\[CLG09\]](#) studied the graphs $\mathfrak{G}_1(p, \ell)$ for primes ℓ , and a significant amount of isogeny-based cryptography is based on these foundations. For $g = 2$, the graph $\mathfrak{G}_2(p, 2)$ is similar to the superspecial $(2, 2)$ -isogeny graph studied by Castryck, Decru and Smith [\[CDS20\]](#). Both of these constructions are aware of the fact that the graph is technically directed, but do not elaborate upon that in a significant way. One could hope to turn this graph into an undirected one due to the existence of dual isogenies, but unfortunately this is impossible. The nontrivial automorphisms on certain abelian varieties create multi-edges which only have one dual, such that several edges would need to share the same “dual edge”. Only in the case of $g = 1$ and $p \equiv 1 \pmod{12}$ are there no nontrivial automorphisms and hence it can be made undirected with ease. Alternatively, one could merge the multi-edges, but this would void the regularity of the graph. We now state some properties of our general digraph.

Consider a walk of length two in this digraph, from $((A_1, H_1), (A_2, H_2))$ to $((A_2, H_2), (A_3, H_3))$ to $((A_3, H_3), (A_4, H_4))$. Then $\Psi_1(H_1) = H_2$ and $\ker \Psi_2 \cap$

$H_2 = 0$ imply that $\Psi_2 \circ \Psi_1$ is an $(\ell^2)^g$ -isogeny. Or, in the terminology of [CDS20], we say that Ψ_2 is a *good extension* of Ψ_1 . Inductively, one can then see that a walk of length n in this digraph corresponds to an $(\ell^n)^g$ -isogeny between two g -dimensional superspecial abelian varieties with an $(\ell)^g$ -level structure. The digraph contains some other nice properties, as can be seen in the following proposition.

Proposition 7. *The superspecial $(\ell)^g$ -isogeny digraph $\mathfrak{G}_g(p, \ell)$ is $\ell^{g(g+1)/2}$ -regular digraph with number of vertices equal to*

$$\mathcal{O}\left((\ell p)^{g(g+1)/2}\right).$$

Proof. The number of superspecial principally polarized abelian varieties over $\mathbb{F}_p$ is $c_g p^{g(g+1)/2}$, with c_g some constant depending on g only [ATY24, §1.1]. By [CD22, Lemma 2], the number of maximal isotropic subgroups for any given abelian variety is

$$\ell^{g(g+1)/2} \cdot \prod_{i=1}^g \left(1 + \frac{1}{\ell^i}\right),$$

which is dominated by $\ell^{g(g+1)/2}$. The locus of (superspecial) principally polarized abelian varieties with extra automorphisms is a proper closed subvariety of the moduli space, hence asymptotically the number of vertices is $\mathcal{O}\left((\ell p)^{g(g+1)/2}\right)$. Lastly, the graph is $\ell^{g(g+1)/2}$ -regular as this is the number of good extensions for any given maximal isotropic subgroup [CD22, Lemma 2]. $\square$

Our main interest is in the expansion properties of superspecial isogeny digraphs. As detailed in Section 2, the eigenvalues of the adjacency matrix A of $\mathfrak{G}_g(p, \ell)$ give us important information on such expansion properties. As we are working with directed graphs, we expect these eigenvalues to be complex numbers $\lambda_i \in \mathbb{C}$. We sort these by their modulus, and, by Proposition 7, we have $\lambda_1 = \ell^{g(g+1)/2} + 0 \cdot i$, where $\ell^{g(g+1)/2}$ is the number of good extensions of degree ℓ . Our main interest for the expansion properties is the modulus and angle of λ_2 , when $p \rightarrow \infty$.

Definition 3. Let A be the adjacency matrix of $\mathfrak{G}_g(p, \ell)$ and let $\lambda_{2,g}(p, \ell)$ be the second largest eigenvalue of A . We define the *general second eigenvalue* $\lambda_{2,g}(\ell)$ as the limit $\lim_{p \rightarrow \infty} \lambda_{2,g}(p, \ell)$, if this converges.

In light of this definition, we will similarly refer to $\lambda_{1,g}(\ell) = \ell^{g(g+1)/2}$ as the general first eigenvalue.

Remark 1. This is in contrast to other isogeny graphs, whose expansion properties can be measured using the *singular values* of the adjacency matrix A , which we find by computing $A^\top A$. We cannot take this approach for superspecial isogeny digraphs: transposing A comes down to inverting edges, but inverting the edge $(A_1, H_1) \rightarrow (A_2, H_2)$ does not make sense for these graphs, as an edge requires that $A_1 = A_2/H_2$ and, in general, the inverse edge $(A_2, H_2) \rightarrow (A_1, H_1)$ will not satisfy $A_2 = A_1/H_1$.

Based on the experimental data for $g = 1$ and $g = 2$, which we detail in [Section 4.2](#), we conjecture the following.

Main Conjecture. Let $\lambda_{2,g}(\ell)$ denote the general second eigenvalue of the $(\ell)^g$ -isogeny digraph $\mathfrak{G}_g(p, \ell)$. Then

$$\lambda_{2,g}(\ell) = \ell^{g^2/2} \in \mathbb{R}.$$

Remark that this implies that the angle of the (complex) second eigenvalue $\lambda_{2,g}(p, \ell)$ converges to zero for $p \rightarrow \infty$. Given that $\lambda_{1,g}$ is equal to the number of good extensions $\ell^{g(g+1)/2}$, the conjecture furthermore implies that the spectral expansion $|\lambda_{2,g}|/|\lambda_{1,g}|$ equals $1/\ell^{g/2}$. Finally, as $\ell^{g^2/2} \leq 2\sqrt{\ell^{g(g+1)/2}} - 1 \approx 2 \cdot \ell^{g(g+1)/4}$, we claim that the superspecial isogeny digraphs $\mathfrak{G}_1(p, \ell)$, $\mathfrak{G}_2(p, 2)$, $\mathfrak{G}_2(p, 3)$ are Ramanujan graphs. More concretely, this implies that taking only *good extensions* strongly improves the expansion properties, in comparison to the general isogeny graphs where bad extensions cause small cycles. As cryptographic use cases often implicitly rely on good extensions, this substantiates the claim that good walks of appropriate length give us a close-to uniform distribution on the codomain.

4.2 Concrete Expansion of Superspecial Isogeny Digraphs

We are interested in concrete expansion properties of superspecial isogeny digraphs in dimension $g = 1$ and $g = 2$ to get an improved understanding of the expansion when considering good extensions.

Dimension 1 In dimension 1, we not only look at ℓ -isogeny graphs, but also N -isogeny graphs for prime powers $N = \ell^k$. We build the N -isogeny digraph $\mathfrak{G}_1(p, N)$ by walking around using Vélú's formulas. Although this has the disadvantage of being much slower than working with modular polynomials, our need to identify the correct paths and double edges necessitates the use of torsion bases. In particular, for $N = \ell^k$, we easily identify and compute good extensions for a given node (E, G) .

Lemma 5. *Let $\ell \neq p$ be a prime, and $N = \ell^k$. Let $(E, G) \in \mathfrak{G}_1(p, N)$ with $G = \langle Q \rangle$ for some $Q \in E[N]$ and $P \in E[N] \setminus \langle Q \rangle$. Then, for every $a \in \mathbb{Z}/N\mathbb{Z}$, the isogeny $\varphi : E \rightarrow E'$ with $\ker \varphi = \langle P + [a]Q \rangle$ is a good extension from (E, G) to $(E', \varphi(Q))$. Furthermore, all N good extensions of (E, G) are of this form.*

Proof. Of the $\ell^k \cdot (\ell + 1)$ isogenies of degree N with domain E , to ensure non-zero overlap with $\langle Q \rangle$, the kernel must have a generator of the form $P + [a]Q$. We find N such isogenies $\varphi : E \rightarrow E'$. The level structure G' on E' is such that $\varphi' : E' \rightarrow E'/G$ is the dual of φ , and is thus given by $G' = \langle \varphi(Q) \rangle$, as this ensures $[N] = \varphi' \circ \varphi$. $\square$

An immediate consequence is that for N prime, every isogeny from (E, G) is a good extension, other than the isogeny with kernel G . Thus, for N prime, we may also work with modular polynomials instead of torsion bases to walk around $\mathfrak{G}_1(p, N)$.

Computations. We have computed $\lambda_2(p, N)$ for all prime powers N up to 32, for primes p up to roughly 2^{14} .

Modulus. We expect $N^{g/2} = \sqrt{N}$ as the size of the second eigenvalue. In [Figure 1](#), we show the normalized value $|\lambda_2(p, N)|/\sqrt{N}$ for all prime powers $N \leq 32$. For primes $N = \ell$, the convergence of the modulus $|\lambda_2(\ell)|$ to $\sqrt{\ell}$ is essentially immediate, hence, it is visible only as the horizontal line with value 1. For composite $N = \ell^k$, the value k has a large impact on the rate of convergence: we see a far slower convergence for larger k .

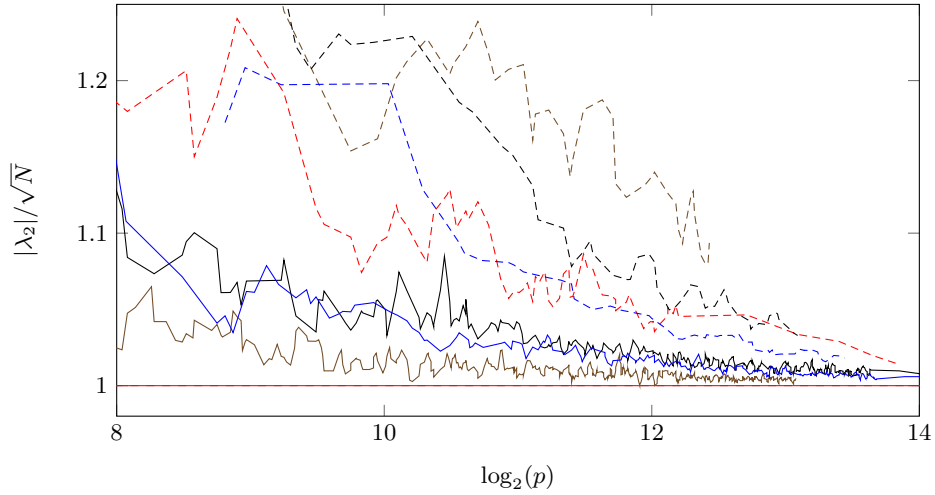


Fig. 1. Convergence of normalized $|\lambda_2|$, for prime powers $N \leq 32$

Angle. The angle converges far slower to 0, as is visible in [Figure 2](#). We believe that this is due to the existence of small cycles in the graph, which arise from endomorphisms with a relatively small degree. For example, the action of the class group on the set of $\mathbb{F}_p$ -rational supersingular curves creates a cycle of roughly size $h(-p) \approx \sqrt{p}$. The existence of small cycles in isogeny graphs has been studied thoroughly, see [\[ABCG+26, ACLS+24, BCEM+19, Gha24, LMOS+25, Orv25\]](#).

The interpretation for a small angle of λ_2 is as follows. Normalizing the adjacency matrix of an m -regular digraph with a factor $1/m$ yields the transition matrix of a Markov process. The dominant eigenvalue of this normalized matrix is λ_1/m and equals one, corresponding to the stationary distribution. The second eigenvalue is $\lambda_2/m = a + bi \in \mathbb{C}$ with accompanying eigenvector v_2 . The angle $\theta = \tan^{-1}(b/a)$ measures the strength of directed cyclic motion (i.e. oscillatory behavior) in that eigenmode. If $|b| \ll |a|$, then $|\theta| \approx 0$ and v_2 's oscillation in time lacks significant rotational component, hence the directed cycles in the

graph responsible for this eigenvalue are almost balanced in forward/backward directions in the support of that eigenvector. The graph thus lacks coherent directed cycles of length related to the phase of this eigenmode (or they exist but are damped heavily by diffusion to other parts). The main dynamic is thus exponential convergence dominated by a , with negligible cyclic behavior, similar to an undirected graph.

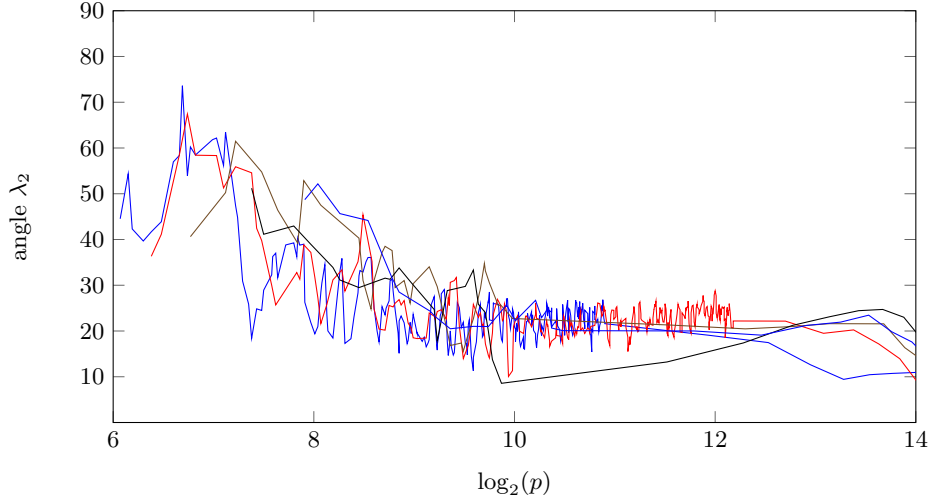


Fig. 2. Convergence of angle of λ_2 , for primes $\ell = 2, 3, 5, 7, 11$

Dimension 2 In dimension 2, we explore the isogeny digraphs $\mathfrak{G}_2(\ell, p)$ for $\ell = 2$ using Richelot isogenies [Ric37, Smi05].

The digraph for $\ell = 2$. When $\ell = 2$, we get $\ell^3 = 8$ good extensions of degree (ℓ, ℓ) . In this subsection, nodes are either Jacobians J or elliptic products $E \times E'$ with a specified isotropic $(2, 2)$ -subgroup G . By Lemmas 6 and 7 for $\ell = 2$, we can compute these 8 good extensions given G both for Jacobians and products of elliptic curves.

For products of elliptic curves, the fifteen $(2, 2)$ -isogenies are divided into 9 diagonal isogenies and 6 gluing isogenies. Depending on whether G represents a diagonal or a gluing isogeny φ , we get:

- if φ is a diagonal isogeny, then there are 4 diagonal kernels G' and 4 gluing kernels G' with $G \cap G' = 0$;
- if φ is a gluing isogeny, then there are 6 diagonal kernels G' and 2 gluing kernels G' with $G \cap G' = 0$,

We use [EPSV24] to generate a supersingular starting curve E , and start walking from E^2 , keeping track of the level structure.

Remark 2. There is one special case for which the normal formulas for diagonal isogenies or gluing isogenies will fail: when $E = E'$, or more generally $E \cong E'$, the $(2, 2)$ -subgroup $G = \{(P, P) \mid P \in E[2]\}$ is the kernel of the self-dual isogeny[‡]

$$\varphi : E \times E \rightarrow E \times E, \quad (P, Q) \mapsto (P + Q, P - Q).$$

We count this isogeny as a gluing isogeny because of the shape of its kernel.

Computations. We have computed $\mathfrak{G}_2(p, \ell)$ for $\ell = 2$ and primes p up to roughly 2^8 . As Figures 3 and 4 show, both the modulus and angle of $\lambda_2(\ell)$ converge surprisingly quickly. As expected, the modulus converges to $\ell^{g^2/2} = 4$, and the angle goes to zero.

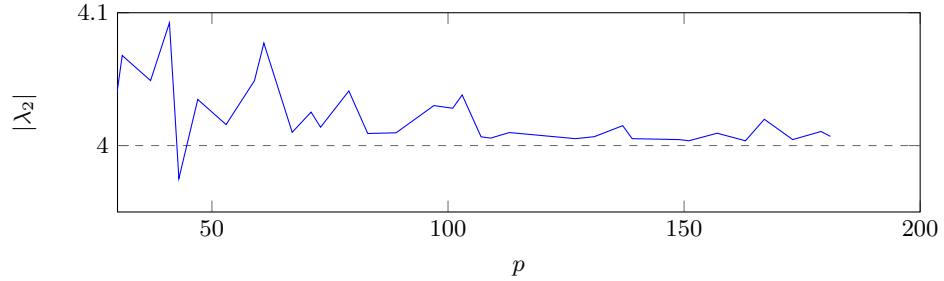


Fig. 3. Convergence of $|\lambda_2|$, for $\ell = 2$

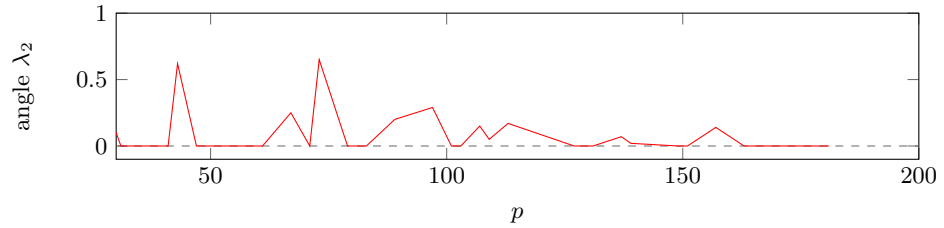


Fig. 4. Convergence of angle of λ_2 , for $\ell = 2$

5 Future work

An evident choice for future work is to see if these heuristic results extend further; i.e. see how the superspecial $(\ell)^g$ -isogeny digraph behaves for $g = 2$

[‡]One may also derive this $(2, 2)$ -isogeny from the Kani diagram where all corners are E and all isogenies the identity.

and $\ell \geq 3$, for $g > 2$ and any ℓ . For $g = 2$ and $\ell = 3$, there are significantly more hurdles to compute $\mathfrak{G}_2(p, 3)$: the formulas we tried [BFT14, DK23] work for almost all possible isotropic $(3, 3)$ -subgroups, apart from some exceptional cases. However, to complete the full graph $\mathfrak{G}_2(p, 3)$ these exceptional cases cause a significant issue. The required $(3, 3)$ -isogeny computations may be sped up a bit by using the formulas from [CCS25] or [DK26]. The latter would require swapping between level-3 theta structures and isomorphism classes of abelian surfaces, however, and in both cases care would need to be taken to handle the exceptional product surfaces. Furthermore, the large number of edges in combination with the relatively low speed of such isogeny computations make the computation of the full graph at the moment unfeasible. More specifically, we cannot work over small primes p , as the proportion of surfaces with non-trivial automorphism group is too large, whereas too large primes p cause infeasibly large adjacency matrices to compute eigenvalues. With the additional restriction $\ell \mid p + 1$ to simplify (ℓ, ℓ) -isogeny computations, this leaves the four primes $p = 41, 47, 53, 59$ as potential candidates to explore $\mathfrak{G}_2(p, 3)$. Remark that even for these p , the number of abelian surfaces with non-trivial automorphism group still vastly outnumber the ones with $\text{RA}(A) = 1$, potentially skewing the results a lot. Using the exact counts from [CDS20, Proposition 2] and [IKO86, Theorem 3.3], this outnumbering is true for all $p \leq 101$. Note that this plays a lesser role for $\ell = 2$. Indeed: apart from potentially one specific curve, all reduced automorphism groups have an order divisible by two and this seems to make the action on the (ℓ, ℓ) -level surfaces “nicer” in some sense. For instance, generic products $E \times E'$ have zero maximal isotropic $(2, 2)$ -subgroups being mapped to one another in the orbit of $[1] \times [-1]$, which is not true for odd ℓ .

For $\mathfrak{G}_2(p, 5)$, the work of Cosset and Robert [CR15] is helpful, but these formulas are noticeably more involved than the rather efficient $(2, 2)$ - and $(3, 3)$ -isogenies. For $\mathfrak{G}_3(p, 2)$, there has been recent work to compute these by means of (multi)radical isogenies in [KMMP+25], although to fully map the graph some nontrivial overhead will come into play, i.e. keeping track of trivially intersecting kernels and swapping from theta structures to isomorphism classes.

Unfortunately, the computational complexity for all these cases quickly explodes due to the fact that there are $\mathcal{O}((\ell p)^{g(g+1)/2})$ vertices in the graph and an increasing amount of edges as well. Besides walking through the entire graph, it also means computing complex eigenvalues of square matrices of dimension 2^{16} or larger. Even for sparse matrices for which we only want λ_2 (which furthermore has a significant gap with λ_1), more specialized methods start failing. For $g > 2$, it also holds that we have many more potential nontrivial reduced automorphism groups, taking up an exceptional proportion of all superspecial abelian varieties for small p . Indeed, the automorphism group of a plane quartic can have an order up to 168. Finally, to actually compute the isogenies, we ideally also have full rational ℓ -torsion to not have to work over too large extension fields, so specific conditions on p should be met. For these reasons, we expect that a theoretical approach, such as the one from [CGL09], may be more fruitful in further outlining the expander properties of these graphs.

References

- [ABCG+26] Sarah Arpin, Ross Bowden, James Clements, Wissam Ghantous, Jason T. LeGrow, and Krystal Maughan. “Cycles and cuts in supersingular L-isogeny graphs”. In: *Finite Fields and Their Applications* 111 (2026), p. 102768. DOI: [10.1016/j.ffa.2025.102768](https://doi.org/10.1016/j.ffa.2025.102768) (cit. on p. 27).
- [ACLS+24] Sarah Arpin, Mingjie Chen, Kristin E. Lauter, Renate Scheidler, Katherine E. Stange, and Ha T. N. Tran. “Orientations and Cycles in Supersingular Isogeny Graphs”. In: *Research Directions in Number Theory: Women in Numbers V*. 2024, pp. 25–86. DOI: [10.1007/978-3-031-51677-1_2](https://doi.org/10.1007/978-3-031-51677-1_2) (cit. on p. 27).
- [ATY24] Yusuke Aikawa, Ryokichi Tanaka, and Takuya Yamauchi. “Isogeny graphs on superspecial abelian varieties: eigenvalues and connection to Bruhat–Tits buildings”. In: *Canadian Journal of Mathematics* 76.6 (2024), pp. 1891–1916. DOI: [10.4153/S0008414X23000676](https://doi.org/10.4153/S0008414X23000676) (cit. on p. 25).
- [BBCS+25] Andrea Basso, Giacomo Borin, Wouter Castryck, Maria Corte-Real Santos, Riccardo Invernizzi, Antonin Leroux, Luciano Maino, Frederik Vercauteren, and Benjamin Wesolowski. “PRISM: Simple and Compact Identification and Signatures from Large Prime Degree Isogenies”. In: *PKC 2025 Proceedings, Part III*. Vol. 15676. Lecture Notes in Computer Science. 2025, pp. 300–332. DOI: [10.1007/978-3-031-91826-1_10](https://doi.org/10.1007/978-3-031-91826-1_10) (cit. on p. 2).
- [BCCF+23] Andrea Basso, Giulio Codogni, Deirdre Connolly, Luca De Feo, Tako Boris Fouotsa, Guido Maria Lido, Travis Morrison, Lorenz Panny, Sikhar Patranabis, and Benjamin Wesolowski. “Supersingular Curves You Can Trust”. In: *EUROCRYPT 2023 Proceedings, Part II*. Vol. 14005. Lecture Notes in Computer Science. 2023, pp. 405–437. DOI: [10.1007/978-3-031-30617-4_14](https://doi.org/10.1007/978-3-031-30617-4_14) (cit. on p. 1).
- [BCEM+19] Efrat Bank, Catalina Camacho-Navarro, Kirsten Eisenträger, Travis Morrison, and Jennifer Park. “Cycles in the Supersingular ℓ -Isogeny Graph and Corresponding Endomorphisms”. In: *Research Directions in Number Theory*. 2019, pp. 41–66. DOI: [10.1007/978-3-030-19478-9_2](https://doi.org/10.1007/978-3-030-19478-9_2) (cit. on p. 27).
- [BDFL+24] Andrea Basso, Pierrick Dartois, Luca De Feo, Antonin Leroux, Luciano Maino, Giacomo Pope, Damien Robert, and Benjamin Wesolowski. “SQIsign2D-West - The Fast, the Small, and the Safer”. In: *ASIACRYPT 2024 Proceedings, Part III*. Vol. 15486. Lecture Notes in Computer Science. 2024, pp. 339–370. DOI: [10.1007/978-981-96-0891-1_11](https://doi.org/10.1007/978-981-96-0891-1_11) (cit. on p. 2).
- [BFT14] Nils Bruin, E. Victor Flynn, and Damiano Testa. “Descent via (3,3)-isogeny on Jacobians of genus 2 curves”. eng. In: *Acta Arith-*

- metica* 165.3 (2014). <http://eudml.org/doc/279018>, pp. 201–223 (cit. on p. 30).
- [CCS25] Maria Corte-Real Santos, Craig Costello, and Benjamin Smith. “Efficient $(3, 3)$ -isogenies on fast Kummer surfaces”. In: *Research in Number Theory* 11.1 (2025), p. 25. DOI: [10.1007/s40993-024-00600-y](https://doi.org/10.1007/s40993-024-00600-y) (cit. on p. 30).
- [CD22] Wouter Castryck and Thomas Decru. “Multiradical isogenies”. In: *Arithmetic, Geometry, Cryptography, and Coding theory 2021*. Vol. 779. Contemporary mathematics. 2022, 57–89. DOI: [http://doi.org/10.1090/conm/779](https://doi.org/10.1090/conm/779) (cit. on pp. 8, 25).
- [CD23] Wouter Castryck and Thomas Decru. “An Efficient Key Recovery Attack on SIDH”. In: *EUROCRYPT 2023 Proceedings, Part V*. Vol. 14008. Lecture Notes in Computer Science. 2023, pp. 423–447. DOI: [10.1007/978-3-031-30589-4_15](https://doi.org/10.1007/978-3-031-30589-4_15) (cit. on p. 2).
- [CDKL+25] Wouter Castryck, Thomas Decru, Péter Kutas, Abel Laval, Christophe Petit, and Yan Bo Ti. “KLPT²: Algebraic Pathfinding in Dimension Two and Applications”. In: *CRYPTO 2025 Proceedings, Part I*. Vol. 16000. Lecture Notes in Computer Science. 2025, pp. 167–200. DOI: [10.1007/978-3-032-01855-7_6](https://doi.org/10.1007/978-3-032-01855-7_6) (cit. on p. 2).
- [CDS20] Wouter Castryck, Thomas Decru, and Benjamin Smith. “Hash functions from superspecial genus-2 curves using Richelot isogenies”. In: *J. Math. Cryptol.* 14.1 (2020), pp. 268–292. DOI: [10.1515/JMC-2019-0021](https://doi.org/10.1515/JMC-2019-0021) (cit. on pp. 2, 6, 8, 24, 25, 30).
- [CGL09] Denis X. Charles, Eyal Z. Goren, and Kristin E. Lauter. “Families of Ramanujan Graphs and Quaternion Algebras”. In: *CRM Proceedings and Lecture Notes*. Vol. 47. 2009. DOI: [10.1090/crmp/047](https://doi.org/10.1090/crmp/047) (cit. on pp. 2, 30).
- [CLG09] Denis Xavier Charles, Kristin E. Lauter, and Eyal Z. Goren. “Cryptographic Hash Functions from Expander Graphs”. In: *J. Cryptol.* 22.1 (2009), pp. 93–113. DOI: [10.1007/S00145-007-9002-X](https://doi.org/10.1007/S00145-007-9002-X) (cit. on pp. 1, 24).
- [CR15] Romain Cosset and Damien Robert. “Computing (ℓ, ℓ) -isogenies in polynomial time on Jacobians of genus 2 curves”. In: *Mathematics of Computation* 84.294 (2015), pp. 1953–1975. URL: <http://www.jstor.org/stable/24489183> (cit. on p. 30).
- [DEFM+25] Pierrick Dartois, Jonathan Komada Eriksen, Tako Boris Fouotsa, Arthur Herlédan Le Merdy, Riccardo Invernizzi, Damien Robert, Ryan Rueger, Frederik Vercauteren, and Benjamin Wesolowski. “PEGASIS: Practical Effective Class Group Action using 4-Dimensional Isogenies”. In: *CRYPTO 2025 Proceedings, Part I*. Vol. 16000. Lecture Notes in Computer Science. 2025, pp. 67–99. DOI: [10.1007/978-3-032-01855-7_3](https://doi.org/10.1007/978-3-032-01855-7_3) (cit. on p. 2).
- [DF24] Max Duparc and Tako Boris Fouotsa. “SQIPrime: A Dimension 2 Variant of SQISignHD with Non-smooth Challenge Isogenies”.

- In: *ASIACRYPT 2024 Proceedings, Part III*. Vol. 15486. Lecture Notes in Computer Science. 2024, pp. 396–429. DOI: [10.1007/978-981-96-0891-1_13](https://doi.org/10.1007/978-981-96-0891-1_13) (cit. on p. 2).
- [DK23] Thomas Decru and Sabrina Kunzweiler. “Efficient Computation of $(3^n, 3^n)$ -Isogenies”. In: *AFRICACRYPT 2023 Proceedings*. Vol. 14064. Lecture Notes in Computer Science. 2023, pp. 53–78. DOI: [10.1007/978-3-031-37679-5_3](https://doi.org/10.1007/978-3-031-37679-5_3) (cit. on p. 30).
- [DK26] Thomas Decru and Sabrina Kunzweiler. *Abelian surfaces in Hesse form and explicit isogeny formulas*. Cryptology ePrint Archive, Paper 2026/039. 2026. URL: <https://eprint.iacr.org/2026/039> (cit. on p. 30).
- [DLRW24] Pierrick Dartois, Antonin Leroux, Damien Robert, and Benjamin Wesolowski. “SQIsignHD: New Dimensions in Cryptography”. In: *EUROCRYPT 2024 Proceedings, Part I*. Vol. 14651. Lecture Notes in Computer Science. 2024, pp. 3–32. DOI: [10.1007/978-3-031-58716-0_1](https://doi.org/10.1007/978-3-031-58716-0_1) (cit. on p. 2).
- [DM] Holger Dell and Dieter van Melkebeek. *Pseudorandomness and Derandomization, Lecture 9: Expanders*. <https://pages.cs.wisc.edu/~dieter/Courses/2013s-CS880/Scribes/PDF/lecture09.pdf> (cit. on p. 10).
- [EHLM+18] Kirsten Eisenträger, Sean Hallgren, Kristin E. Lauter, Travis Morrison, and Christophe Petit. “Supersingular Isogeny Graphs and Endomorphism Rings: Reductions and Solutions”. In: *EUROCRYPT 2018 Proceedings, Part III*. Vol. 10822. Lecture Notes in Computer Science. 2018, pp. 329–368. DOI: [10.1007/978-3-319-78372-7_11](https://doi.org/10.1007/978-3-319-78372-7_11) (cit. on p. 1).
- [EPSV24] Jonathan Komada Eriksen, Lorenz Panny, Jana Sotáková, and Mattia Veroni. “Deuring for the People: Supersingular Elliptic Curves with Prescribed Endomorphism Ring in General Characteristic”. In: *LuCaNT: LMFDB, Computation, and Number Theory*. Vol. 796. Contemporary mathematics. 2024, pp. 339–373. DOI: [10.1090/conm/796](https://doi.org/10.1090/conm/796) (cit. on p. 28).
- [FKLP+20] Luca De Feo, David Kohel, Antonin Leroux, Christophe Petit, and Benjamin Wesolowski. “SQISign: Compact Post-quantum Signatures from Quaternions and Isogenies”. In: *ASIACRYPT 2020 Proceedings, Part I*. Ed. by Shiho Moriai and Huaxiong Wang. Vol. 12491. Lecture Notes in Computer Science. 2020, pp. 64–93. DOI: [10.1007/978-3-030-64837-4_3](https://doi.org/10.1007/978-3-030-64837-4_3) (cit. on p. 2).
- [FS21] Enric Florit and Benjamin Smith. *An atlas of the Richelot isogeny graph*. Cryptology ePrint Archive, Paper 2021/013. 2021. URL: <https://eprint.iacr.org/2021/013> (cit. on pp. 10, 12, 14, 17, 20, 23, 37, 39–41, 43).
- [FS22] Enric Florit and Benjamin Smith. “Automorphisms and isogeny graphs of abelian varieties, with applications to the superspecial Richelot isogeny graph”. In: *Arithmetic, Geometry, Cryptogra-*

- phy, and Coding theory 2021*. Vol. 779. Contemporary mathematics. 2022, 103–132. DOI: <http://doi.org/10.1090/conm/779> (cit. on p. 2).
- [FT19] E. Victor Flynn and Yan Bo Ti. “Genus Two Isogeny Cryptography”. In: *PQCrypto 2019 Proceedings*. Vol. 11505. Lecture Notes in Computer Science. 2019, pp. 286–306. DOI: [10.1007/978-3-030-25510-7_16](https://doi.org/10.1007/978-3-030-25510-7_16) (cit. on p. 2).
- [Gha24] Wissam Gbantous. “Loops, multi-edges and collisions in supersingular isogeny graphs”. In: *Advances in Mathematics of Communications* 18.4 (2024), pp. 935–955. DOI: [10.3934/amc.2022038](https://doi.org/10.3934/amc.2022038) (cit. on p. 27).
- [IKO86] Tomoyoshi Ibukiyama, Toshiyuki Katsura, and Frans Oort. “Supersingular curves of genus two and class numbers”. In: *Compositio Mathematica* 57.2 (1986), pp. 127–152. URL: <http://eudml.org/doc/89752> (cit. on pp. 17, 19, 21, 30).
- [JF11] David Jao and Luca De Feo. “Towards Quantum-Resistant Cryptosystems from Supersingular Elliptic Curve Isogenies”. In: *PQCrypto 2011 Proceedings*. Vol. 7071. Lecture Notes in Computer Science. 2011, pp. 19–34. DOI: [10.1007/978-3-642-25405-5_2](https://doi.org/10.1007/978-3-642-25405-5_2) (cit. on p. 2).
- [JZ25] Bruce W. Jordan and Yevgeny Zaytman. *Isogeny graphs of superspecial abelian varieties and Brandt matrices*. 2025. URL: <https://arxiv.org/abs/2005.09031> (cit. on p. 2).
- [Kan97] Ernst Kani. “The number of curves of genus two with elliptic differentials.” In: *Journal für die reine und angewandte Mathematik* 1997.485 (1997), pp. 93–122. DOI: [doi:10.1515/crll.1997.485.93](https://doi.org/10.1515/crll.1997.485.93) (cit. on p. 2).
- [KLPT14] David Kohel, Kristin E. Lauter, Christophe Petit, and Jean-Pierre Tignol. “On the quaternion -isogeny path problem”. In: *LMS J. Comput. Math.* 17.Theory (2014), pp. 418–432. DOI: [10.1112/S1461157014000151](https://doi.org/10.1112/S1461157014000151) (cit. on p. 1).
- [KMMP+25] Sabrina Kunzweiler, Luciano Maino, Tomoki Moriya, Christophe Petit, Giacomo Pope, Damien Robert, Miha Stopar, and Yan Bo Ti. “Radical 2-Isogenies and Cryptographic Hash Functions in Dimensions 1, 2 and 3”. In: *PKC 2025 Proceedings, Part III*. Vol. 15676. Lecture Notes in Computer Science. 2025, pp. 265–299. DOI: [10.1007/978-3-031-91826-1_9](https://doi.org/10.1007/978-3-031-91826-1_9) (cit. on p. 30).
- [KTW22] Sabrina Kunzweiler, Yan Bo Ti, and Charlotte Weitkämper. “Secret Keys in Genus-2 SIDH”. In: *SAC 2021: 28th Annual International Workshop on Selected Areas in Cryptography*. Vol. 13203. Lecture Notes in Computer Science. 2022, pp. 483–507. DOI: [10.1007/978-3-030-99277-4_23](https://doi.org/10.1007/978-3-030-99277-4_23) (cit. on pp. 23, 47).
- [LMOS+25] Jun Bo Lau, Travis Morrison, Eli Orvis, Gabrielle Scullard, and Lukas Zobernig. *Zeta functions of abstract isogeny graphs and*

- modular curves*. 2025. URL: <https://arxiv.org/abs/2509.15214> (cit. on p. 27).
- [MMPP+23] Luciano Maino, Chloe Martindale, Lorenz Panny, Giacomo Pope, and Benjamin Wesolowski. “A Direct Key Recovery Attack on SIDH”. In: *EUROCRYPT 2023 Proceedings, Part V*. Vol. 14008. Lecture Notes in Computer Science. 2023, pp. 448–471. DOI: [10.1007/978-3-031-30589-4_16](https://doi.org/10.1007/978-3-031-30589-4_16) (cit. on p. 2).
- [NOCC+24] Kohei Nakagawa, Hiroshi Onuki, Wouter Castryck, Mingjie Chen, Riccardo Invernizzi, Gioella Lorenzon, and Frederik Vercauteren. “SQIsign2D-East: A New Signature Scheme Using 2-Dimensional Isogenies”. In: *ASIACRYPT 2024 Proceedings, Part III*. Vol. 15486. Lecture Notes in Computer Science. 2024, pp. 272–303. DOI: [10.1007/978-981-96-0891-1_9](https://doi.org/10.1007/978-981-96-0891-1_9) (cit. on p. 2).
- [Orv25] Eli Orvis. “Distribution of cycles in supersingular ℓ -isogeny graphs”. In: *Journal of Number Theory* 277 (2025), pp. 236–261. DOI: <https://doi.org/10.1016/j.jnt.2025.03.013> (cit. on p. 27).
- [Ric37] Friedrich Julius Richelot. “De transformatione integralium Abelianorum primi ordinis commentatio. Caput secundum. De computatione integralium Abelianorum primi ordinis.” In: *Journal für die reine und angewandte Mathematik* 1837.16 (1837), pp. 285–341. DOI: [doi:10.1515/crll.1837.16.285](https://doi.org/10.1515/crll.1837.16.285) (cit. on pp. 8, 28).
- [Rob23] Damien Robert. “Breaking SIDH in Polynomial Time”. In: *EUROCRYPT 2023 Proceedings, Part V*. Vol. 14008. Lecture Notes in Computer Science. 2023, pp. 472–503. DOI: [10.1007/978-3-031-30589-4_17](https://doi.org/10.1007/978-3-031-30589-4_17) (cit. on p. 2).
- [Rob24] Damien Robert. *The module action for isogeny based cryptography*. Cryptology ePrint Archive, Paper 2024/1556. 2024. URL: <https://eprint.iacr.org/2024/1556> (cit. on p. 2).
- [Smi05] Benjamin Smith. “Explicit endomorphisms and correspondences”. PhD thesis. University of Sydney, 2005 (cit. on pp. 8, 28).
- [Tak18] Katsuyuki Takashima. “Efficient Algorithms for Isogeny Sequences and Their Cryptographic Applications”. In: *Mathematical Modelling for Next-Generation Cryptography: CREST Crypto-Math Project*. 2018, pp. 97–114. DOI: [10.1007/978-981-10-5065-7_6](https://doi.org/10.1007/978-981-10-5065-7_6) (cit. on p. 2).
- [The24] The Sage Developers. *SageMath, the Sage Mathematics Software System (Version 10.4)*. <https://www.sagemath.org>. 2024 (cit. on p. 5).
- [Vad] Salil P. Vadhan. *Harvard lecture notes, Pseudorandomness I, Chapter 4: Expander graphs*. <https://people.seas.harvard.edu/~salil/cs225/spring09/lecnotes/Chap4.pdf> (cit. on p. 10).

- [Vél71] Jacques Vélú. “Isogénies entre courbes elliptiques”. In: *Comptes-Rendus de l’Académie des Sciences* 273 (1971), pp. 238–241 (cit. on p. [8](#)).

A Automorphisms of Curves and Jacobians

Florit and Smith [FS21] give a detailed atlas of the nontrivial automorphisms in genus 2, classified by the shape of the hyperelliptic curve. Whereas Florit and Smith mainly use Bolza’s names in their classification, we will strictly refer to Jacobians with respect to a certain reduced automorphism group. We mainly do so to avoid confusion: the naming in type system has no relation to the order of the automorphisms, which makes it harder to remember which type refers to which automorphism.

Jacobians. Figure 6 shows the 7 possible groups $\text{RA}(J)$. Beyond the 3-dimensional moduli space of generic Jacobians with $\text{RA}(J) = 1$, nontrivial automorphisms only appear in specific situations. All Jacobians where $\text{RA}(J)$ has a $\mathbb{Z}/2\mathbb{Z}$ -component are $(2, 2)$ -isogenous to a product of elliptic curves $E \times E'$, which gives a 2-dimensional subspace. The 1-dimensional moduli subspaces are given by restrictions between E and E' . If $E = E'$ and E has no nontrivial automorphisms, the $(2, 2)$ -isogenous Jacobians have $\text{RA}(J) = (\mathbb{Z}/2\mathbb{Z})^2$, and if E and E' are 3-isogenous by φ , the Jacobians with $\text{RA}(J) = \mathcal{S}_3$ are given by $J = (E \times E')/\Gamma$ where $\Gamma = \{(P, \varphi(P)) \mid P \in E[2]\}$.[§]

The 0-dimensional moduli subspaces are all unique up to isomorphism. The only Jacobian J with $\text{RA}(J) \cong \mathcal{D}_{2 \times 6}$ is $(2, 2)$ -isogenous to $E_0 \times E_0$, and is the Jacobian of the curve $C : y^2 = x^6 + 1$. The only Jacobian J with $\text{RA}(J) = \mathcal{S}_4$ is the Jacobian of the curve $C : y^2 = x^5 - x$, and is $(2, 2)$ -isogenous to the exceptional curve[¶] $E \times E$ where $j(E) = 8000$. Lastly, the Jacobian J with $\text{RA}(J) \cong \mathbb{Z}/5\mathbb{Z}$ is *not* $(2, 2)$ -isogenous to a product of curves; it is the Jacobian of the curve $C : y^2 = x^5 - 1$.

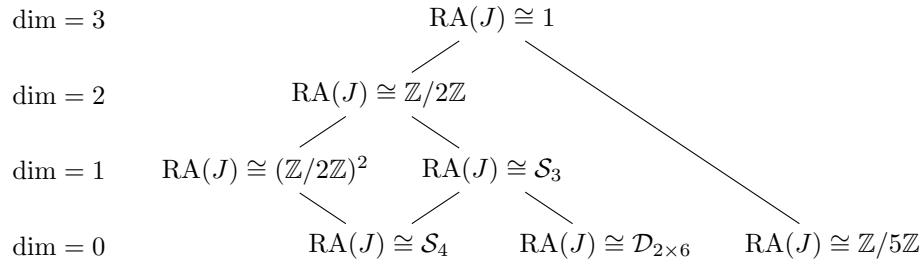


Fig. 5. A direct copy of Figure 1 from [FS21], showing the possible reduced automorphism groups of 2-dimensional Jacobians. Lines indicate subtypes.

[§]This unique behavior can be explained by the Kani diagram with factorizations $E \xrightarrow{\varphi} E' \rightarrow E'$ and $E \rightarrow E \xrightarrow{\varphi} E'$, which implies that $E \times E'$ has a $(4, 4)$ -endomorphism that splits via J .

[¶]These requirements force E to have a degree-3 endomorphism, making this a specialized case of the two cases above: we have both $E = E'$ and a 3-isogeny $\varphi : E \rightarrow E'$.

These reduced automorphism groups can be described with only a few automorphisms, because subtypes inherit automorphisms from their supertypes. We use the greek letter to denote the automorphism on (the x -coordinate of) the curve C , and the subscript $*$ to denote the action on elements of the Jacobian $J = \text{Jac}(C)$. We give an overview of Jacobians and their reduced automorphism groups in [Table 5](#), with the specific automorphisms described in [Table 4](#).

Table 3. Possible reduced automorphism groups of Jacobians

RA(J)	Dimension	(2,2)-isogenous $E \times E'$	Generators
1	3	-	-
$\mathbb{Z}/2\mathbb{Z}$	2	generic E, E'	σ_*
$(\mathbb{Z}/2\mathbb{Z})^2$	1	isomorphic $E \rightarrow E'$	σ_*, τ_*
S_3	1	3-isogenous $E \rightarrow E'$	σ_*, ρ_*
S_4	0	E with 3-endomorphism	ρ_*, ω_*
$\mathcal{D}_{2 \times 6}$	0	$E_0 \times E_0$	$\tau_*, \zeta_*^{(6)}$
$\mathbb{Z}/5\mathbb{Z}$	0	-	$\zeta_*^{(5)}$

Table 4. Automorphisms on specific types of Jacobians

Autom.	On C	Order	Conditions on Curve	Eq.
σ	$x \mapsto -x$	2	$y^2 = (x^2 - 1)(x^2 - s)(x^2 - t)$	
τ	$x \mapsto 1/x$	2	$y^2 = (x^2 - 1)(x^2 - u^2)(x^2 - \frac{1}{u^2})$	
ρ	See Eq. (1)	3	$y^2 = (x^2 - 1)(x^2 - s_v)(x^2 - t_v)$	
$\zeta^{(6)}$	$x \mapsto \zeta_6 x$	6	$y^2 = x^6 + 1$	
ω	See Eq. (2)	4	$y^2 = x^5 - x$ [¶]	
$\zeta^{(5)}$	$x \mapsto \zeta_5 x$	5	$y^2 = x^5 - 1$	

$$\rho : x \mapsto \frac{(2\zeta_3 + 1)(v^2 - 1)x + 3(v + 1)^2}{3(v - 1)^2x + (2\zeta_3 + 1)(v^2 - 1)}, \quad (1)$$

$$s_v := \frac{(v + 1)(v - \zeta_3)}{(v - 1)(v + \zeta_3)}, \quad t_v := \frac{(v + 1)(v - \zeta_3^2)}{(v - 1)(v + \zeta_3^2)}$$

$$\omega : x \mapsto \frac{x - (\sqrt{2} + 1)}{(\sqrt{2} - 1)x + 1} \quad (2)$$

[¶]The automorphism ω from [Eq. \(2\)](#) acts on the isomorphic curve $y^2 = (x^2 - 1)(x^2 - s_v)(x^2 - t_v)$ following [Eq. \(1\)](#) with $v = (\zeta_{12}^2 + \zeta_{12} + 1)/\sqrt{2}$.

Products of Elliptic Curves. In the case of elliptic curves, there are seven possible reduced automorphism groups, where most nontrivial automorphisms are derived from the nontrivial automorphisms on E_0 and E_{1728} .

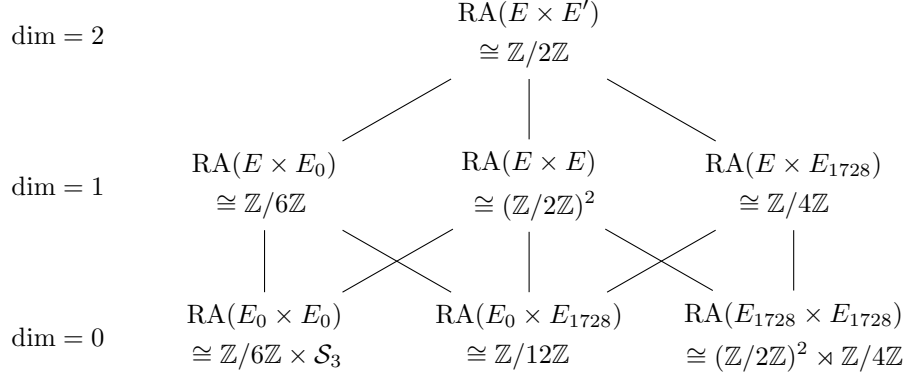


Fig. 6. A direct copy of Figure 2 from [FS21], showing the possible reduced automorphism groups of products of elliptic curves. Lines indicate subtypes.

See Table 6 for the specific automorphisms on elliptic products.

Table 5. Possible reduced automorphism groups of products of elliptic curves

$\text{RA}(E \times E')$	Dimension	Properties	Generators
$\mathbb{Z}/2\mathbb{Z}$	2	generic E, E'	κ
$(\mathbb{Z}/2\mathbb{Z})^2$	1	isomorphic $E \cong E'$	κ, χ
$\mathbb{Z}/6\mathbb{Z}$	1	$E' \cong E_0$	$[1] \times -\zeta^{(3)}$
$\mathbb{Z}/4\mathbb{Z}$	1	$E' \cong E_{1728}$	$[1] \times \iota$
$\mathbb{Z}/12\mathbb{Z}$	0	$E \cong E_{1728}, E' \cong E_0$	$\iota \times \zeta^{(3)}$
$(\mathbb{Z}/2\mathbb{Z})^2 \rtimes \mathbb{Z}/4\mathbb{Z}$	0	$E \cong E' \cong E_{1728}$	$\chi, [1] \times \iota$
$\mathbb{Z}/6\mathbb{Z} \times \mathcal{S}_3$	0	$E \cong E' \cong E_0$	$\chi, [1] \times -\zeta^{(3)}$

B Reduced automorphisms on products of elliptic curves

Remark that in the standard notation, a symplectic basis for $(E \times E')[\ell]$ is of the form $((P, \infty), (\infty, P'), (Q, \infty), (\infty, Q'))$. Ergo, when discussing matrix representations of automorphisms, the upper-left 2×2 block does *not* represent the $E[\ell]$ component nor does the lower-right 2×2 block represent the $E'[\ell]$ component.

Table 6. Automorphisms on products of elliptic curves

Autom.	Action on $E \times E'$	Order	Types
κ	$(P, Q) \mapsto (P, -Q)$	2	-
χ	$(P, Q) \mapsto (Q, P)$	2	$E \cong E'$
$[1] \times \zeta^{(3)}$	$(P, Q) \mapsto (P, (\zeta_3 x_Q, y_Q))$	3	$E' \cong E_0$
$[1] \times \iota$	$(P, Q) \mapsto (P, (-x_Q, iy_Q))$	4	$E' \cong E_{1728}$

B.1 $\text{RA}(E \times E') = \mathbb{Z}/2\mathbb{Z}$

This is the generic case stemming from the automorphism $\kappa = [1] \times [-1]$, where $E \not\cong E'$ and neither E nor E' having j -invariant 0 or 1728.

Proposition 8. *Let $E \times E'$ be a product of elliptic curves over a field of characteristic $p > 5$, with the automorphism $\kappa = [1] \times [-1]$ of order 2, and $\ell \neq p$ a prime. Then κ acts on the (ℓ, ℓ) -level surfaces of $E \times E'$ as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2
$\ell = 2$	15	0
$\ell \neq 2$	$(\ell + 1)^2$	$\ell^3 - \ell$

Proof. (i) For $\ell = 2$, see [FS21, Subsection 4.4].

(ii) For $\ell \neq 2$, it is easy to see that κ acts as $\text{diag}(1, -1, 1, -1) \in \text{Sp}_4(\mathbb{F}_\ell)$ and the result from Proposition 1(ii) also holds. □

This immediately leads to the following result.

Theorem 7. *Let $E \times E'$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\text{RA}(E \times E') = \mathbb{Z}/2\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E \times E', G)$ is equal to 15 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{2}(\ell^3 + 2\ell^2 + 3\ell + 2).$$

Proof. This is a direct consequence of Proposition 8. □

B.2 $\text{RA}(E \times E') = (\mathbb{Z}/2\mathbb{Z})^2$

This is the case when $E \cong E'$, and $j(E) \notin \{0, 1728\}$. The additional automorphism of order two is given by $\chi : (P, Q) \mapsto (Q, P)$, assuming $E' = E$.

Proposition 9. *Let $E \times E$ be a product of elliptic curves over a field of characteristic $p > 5$, with the automorphism $\chi : (P, Q) \mapsto (Q, P)$ of order 2, and $\ell \neq p$ a prime. Then χ acts on the (ℓ, ℓ) -level surfaces of $E \times E$ as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2
$\ell = 2$	7	8
$\ell \neq 2$	$(\ell + 1)^2$	$\ell^3 - \ell$

Proof. (i) For $\ell = 2$, see [FS21, Subsection 4.6].
(ii) For $\ell \neq 2$, it is easy to see that χ acts as

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell),$$

but with a symplectic change of basis to $((P, P), (P, -P), (Q, Q), (Q, -Q))$ we see that it is symplectically similar to $\mathrm{diag}(1, -1, 1, -1) \in \mathrm{Sp}_4(\mathbb{F}_\ell)$ and the same result as Proposition 8(ii) holds. $\square$

Combining Propositions 8 and 9, we get the following result.

Theorem 8. *Let $E \times E$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\mathrm{RA}(E \times E) = (\mathbb{Z}/2\mathbb{Z})^2$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E \times E, G)$ is equal to 11 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{4} \left(\ell^3 + 3\ell^2 + 6\ell + 5 + \left(\frac{-1}{\ell} \right) \right).$$

Proof. Burnside's lemma says that the number of (ℓ, ℓ) -level surfaces is equal to

$$\frac{1}{|(\mathbb{Z}/2\mathbb{Z})^2|} \cdot (|X^{\mathrm{id}}| + |X^\kappa| + |X^\chi| + |X^{\kappa\chi}|),$$

hence by Propositions 8 and 9, it suffices to check $X^{\kappa\chi}$. One can verify that the minimal polynomial of $\kappa\chi$ is $x^2 + 1$, so the orbits act exactly the same as τ_* from Proposition 2. $\square$

B.3 $\mathrm{RA}(E \times E') = \mathbb{Z}/6\mathbb{Z}$

This is the case when $j(E) \notin \{0, 1728\}$ and $E' \cong E_0$. The additional automorphism of order three is given by $[1] \times \zeta^{(3)}$, where $\zeta^{(3)} : x \mapsto \zeta_3 x$ on E_0 .

Proposition 10. *Let $E \times E_0$ be a product of elliptic curves over a field of characteristic $p > 5$, with $[1] \times \zeta^{(3)}$ an automorphism of order 3, and $\ell \neq p$ a prime. Then $[1] \times \zeta^{(3)}$ acts on the (ℓ, ℓ) -level surfaces of $E \times E_0$ as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 3
$\ell = 3$	4	36
$\ell \equiv 1 \pmod{3}$	$2\ell + 2$	$(\ell + 1)^2(\ell - 1)$
$\ell \equiv 2 \pmod{3}$	0	$\ell^3 + \ell^2 + \ell + 1$

Proof. (i) For $\ell = 3$, $\zeta^{(3)}$ is unipotent of order 3 and $[1] \times \zeta^{(3)}$ is given by

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell).$$

Any line from $E[3]$ can thus be combined with one specific $\langle (P, 0) \rangle \subseteq E_0[3]$ to create a maximal isotropic $[1] \times \zeta^{(3)}$ -invariant subspace.

- (ii) For $\ell \equiv 1 \pmod{3}$, $\zeta^{(3)}$ is diagonalizable so the action of $[1] \times \zeta^{(3)}$ is given by $\mathrm{diag}(1, \mu_3, 1, \mu_3^2) \in \mathrm{Sp}_4(\mathbb{F}_\ell)$. By [Lemma 3](#), we can combine any line from $E[\ell]$ with the one-dimensional subspaces belonging to μ_3 or μ_3^2 . There are thus $2\ell + 2$ maximal isotropic $[1] \times \zeta^{(3)}$ -invariant subspaces.
- (iii) For $\ell \equiv 2 \pmod{3}$, $\zeta^{(3)}$ is not diagonalizable so the action of $[1] \times \zeta^{(3)}$ is given by

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell).$$

The only $[1] \times \zeta^{(3)}$ -invariant two-dimensional subspaces are clearly $E[\ell]$ and $E_0[\ell]$, but those are not maximal isotropic. □

Using κ from [Proposition 8](#), we can conclude with the following.

Theorem 9. *Let $E \times E_0$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\mathrm{RA}(E \times E_0) = \mathbb{Z}/6\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E \times E_0, G)$ is equal to 5 if $\ell = 2$, equal to 12 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{6} \left(\ell^3 + 2\ell^2 + 7\ell + 6 + (4\ell + 4) \left(\frac{\ell}{3} \right) \right).$$

Proof. Burnside's lemma says that the number of (ℓ, ℓ) -level surfaces is equal to

$$\frac{1}{|\mathbb{Z}/6\mathbb{Z}|} \cdot \left(|X^{\mathrm{id}}| + 2|X^{[1] \times \zeta^{(3)}}| + |X^\kappa| + 2|X^{[1] \times -\zeta^{(3)}}| \right).$$

The behaviour of $[1] \times -\zeta^{(3)}$ is easily determined from [Proposition 10](#) since $[-1]$ is explicit. In particular we have that $|X^{[1] \times \zeta^{(3)}}| = |X^{[1] \times -\zeta^{(6)}}|$. The rest follows from [Propositions 8](#) and [10](#). $\square$

B.4 $\mathbf{RA}(E \times E') = \mathbb{Z}/4\mathbb{Z}$

This is the case when $j(E) \notin \{0, 1728\}$ and $E' \cong E_{1728}$. The additional automorphism of order four is given by $[1] \times \iota$, where $\iota : (x, y) \mapsto (-x, iy)$ on E_{1728} , with $\iota^2 = -1$.

Proposition 11. *Let $E \times E_{1728}$ be a product of elliptic curves over a field of characteristic $p > 5$, with $[1] \times \iota$ an automorphism of order 4, and $\ell \neq p$ a prime. Then $[1] \times \iota$ acts on the (ℓ, ℓ) -level surfaces of $E \times E_{1728}$ as follows:*

(ℓ, ℓ) -level surfaces in orbits of size 1		(ℓ, ℓ) -level surfaces in orbits of size 2	(ℓ, ℓ) -level surfaces in orbits of size 4
$\ell = 2$	3	12	0
$\ell \equiv 1 \pmod{4}$	$2\ell + 2$	$\ell^2 - 1$	$\ell^3 - \ell$
$\ell \equiv 3 \pmod{4}$	0	$(\ell + 1)^2$	$\ell^3 - \ell$

Proof. Note that $\iota^2 = \kappa$ so that distinction goes back to [Proposition 8](#). The minimal polynomial of $[1] \times \iota$ is the product of the separate minimal polynomials and hence $(x - 1)(x^2 + 1)$.

- (i) For $\ell = 2$, see [\[FS21, Subsection 4.10\]](#).
- (ii) For $\ell \equiv 1 \pmod{4}$, the minimal polynomial of ι splits in linear factors such that the action of $[1] \times \iota$ is given by $\text{diag}(1, i, 1, -i) \in \text{Sp}_4(\mathbb{F}_\ell)$. By [Lemma 3](#), we can thus choose a line on $E[\ell]$ with the one-dimensional subspaces belonging to i or $-i$. There are $2\ell + 2$ maximal isotropic $[1] \times \iota$ -invariant subspaces.
- (iii) For $\ell \equiv 3 \pmod{4}$, the minimal polynomial of ι does not factor such that the action of $[1] \times \iota$ is given by

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \in \text{Sp}_4(\mathbb{F}_\ell).$$

The subspace $E_0[\ell]$ is clearly irreducible but it's not isotropic, nor is complement $E[\ell]$, so there are no maximal isotropic $[1] \times \iota$ -invariant subspaces. $\square$

This immediately leads to the following result.

Theorem 10. *Let $E \times E_{1728}$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\text{RA}(E \times E_{1728}) = \mathbb{Z}/4\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E \times E_{1728}, G)$ is equal to 9 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{4} \left(\ell^3 + 2\ell^2 + 5\ell + 4 + (2\ell + 2) \left(\frac{-1}{\ell} \right) \right).$$

Proof. It is clear that ι^3 acts the same as ι , so the statement follows from Burnside's lemma and [Propositions 8](#) and [11](#). $\square$

B.5 $\text{RA}(E \times E') = \mathbb{Z}/12\mathbb{Z}$

This is the case when $E \cong E_{1728}$ and $E' \cong E_0$. No additional automorphisms are needed apart from the already discussed $\iota \times [1]$ and $[1] \times \zeta^{(3)}$.

Theorem 11. *Let $E_0 \times E_{1728}$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\text{RA}(E_0 \times E_{1728}) = \mathbb{Z}/12\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E_0 \times E_{1728}, G)$ is equal to 3 if $\ell = 2$, is equal to 6 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{12} \left(\ell^3 + 2\ell^2 + 9\ell + 12 + 2(\ell + 3) \left(\frac{-1}{\ell} \right) + 4(\ell + 2) \left(\frac{\ell}{3} \right) + 4 \left(\frac{3}{\ell} \right) \right).$$

Proof. We have that the number is given by

$$\frac{1}{12} \cdot \left(|X^{\text{id}}| + |X^{\kappa}| + 2|X^{[1] \times \zeta^{(3)}}| + 2|X^{\iota \times [1]}| + 2|X^{[1] \times -\zeta^{(3)}}| + 4|X^{\iota \times \zeta^{(3)}}| \right),$$

since the choice of a root of unity should not change the outcome. By [Propositions 8](#), [10](#) and [11](#), it suffices to verify the fixed elements of the action of $\iota \times \zeta^{(3)}$, which has order 12. For $\ell \equiv 1 \pmod{12}$ this means the action is diagonalizable as $\text{diag}(i, \mu_3, -i, \mu_3^2) \in \text{Sp}_4(\mathbb{F}_\ell)$ and by [Lemma 3](#) we can create 4 maximal isotropic $\iota \times \zeta^{(3)}$ -invariant subspaces from this. For $\ell \equiv 5, 7, 11 \pmod{12}$, either ι or $\zeta^{(3)}$ is not diagonalizable and it yields no fixed subspaces. $\square$

B.6 $\text{RA}(E \times E') = (\mathbb{Z}/2\mathbb{Z})^2 \rtimes \mathbb{Z}/4\mathbb{Z}$

This is the case when $E \cong E' \cong E_{1728}$. No additional automorphisms are needed and the group is generated by $\langle \chi, [1] \times \iota \rangle$.

Theorem 12. *Let $E_{1728} \times E_{1728}$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\text{RA}(E_{1728} \times E_{1728}) = (\mathbb{Z}/2\mathbb{Z})^2 \rtimes \mathbb{Z}/4\mathbb{Z}$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E_{1728} \times E_{1728}, G)$ is equal to 5 if $\ell = 2$ and for $\ell > 2$ it is*

$$\frac{1}{16} \left(\ell^3 + 4\ell^2 + 15\ell + 24 + (4\ell + 12) \left(\frac{-1}{\ell} \right) + 4 \left(\frac{-2}{\ell} \right) \right).$$

Proof. All automorphisms of $E_{1728} \times E_{1728}$ are given by $(P, Q) \mapsto (\iota^a(P), \iota^b(Q))$ or $(P, Q) \mapsto (\iota^c(P), \iota^d(Q))$, where $b, d \in \{0, 1\}$ and $a, c \in \{0, 1, 2, 3\}$ cover all options up to equivalence. One can verify that this group contains

- one identity element id ;
- three elements of order two with minimal polynomial $x^2 - 1$ acting as κ from [Proposition 8](#);
- four elements of order two with minimal polynomial $x^2 + 1$ acting as τ_* from [Proposition 2](#);
- four elements of order four with minimal polynomial $(x - 1)(x^2 + 1)$ acting as $[1] \times \iota$ from [Proposition 11](#);
- four elements of order four with minimal polynomial $x^4 + 1$ acting as ω_* from [Proposition 5](#).

We thus have that the number of (ℓ, ℓ) -level surfaces is given by

$$\frac{1}{16} \cdot \left(|X^{\text{id}}| + 3|X^\kappa| + 4|X^{\tau_*}| + 4|X^{[1] \times \iota}| + 4|X^{\omega_*}| \right).$$

□

B.7 $\text{RA}(E \times E') = \mathbb{Z}/6\mathbb{Z} \times \mathcal{S}_3$

This is the case when $E \cong E' \cong E_0$. No additional automorphisms are needed and the group is generated by $\langle \chi, [1] \times -\zeta^{(3)} \rangle$. Some of these automorphisms - such as $\chi \circ ([1] \times -\zeta^{(3)})$ have nonreduced order twelve however, which we have not had to discuss previously.

Proposition 12. *Let $E_0 \times E_0$ be a product of elliptic curves over a field of odd characteristic $p > 5$, with $\chi \circ ([1] \times -\zeta^{(3)})$ an automorphism of nonreduced order 12, and $\ell \neq p$ a prime. Then $\chi \circ ([1] \times -\zeta^{(3)})$ acts on the (ℓ, ℓ) -level surfaces of $E_0 \times E_0$ as follows:*

	(ℓ, ℓ) -level surfaces in orbits of size 1	(ℓ, ℓ) -level surfaces in orbits of size 2	(ℓ, ℓ) -level surfaces in orbits of size 3	(ℓ, ℓ) -level surfaces in orbits of size 6
$\ell = 2$	1	2	6	6
$\ell = 3$	1	0	3	36
$\ell \equiv 1 \pmod{12}$	4	$\ell - 1$	$\ell - 1$	$\ell^3 + \ell^2 - \ell - 1$
$\ell \equiv 5 \pmod{12}$	2	$\ell - 1$	$\ell + 1$	$\ell^3 + \ell^2 - \ell - 1$
$\ell \equiv 7 \pmod{12}$	2	$\ell + 1$	$\ell - 1$	$\ell^3 + \ell^2 - \ell - 1$
$\ell \equiv 11 \pmod{12}$	0	$\ell + 1$	$\ell + 1$	$\ell^3 + \ell^2 - \ell - 1$

Proof. Since $(\chi \circ ([1] \times -\zeta^{(3)}))^6 = [-1]$, its minimal polynomial is $\Phi_{12} = x^4 - x^2 + 1$. One can multiply the corresponding symplectic generic matrices to obtain the

generic matrix again before taking into account splitting behaviour in specific $\mathbb{F}_\ell$:

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell).$$

The main use for this matrix is that we can compute its square and cube and find the respective minimal polynomials $x^2 - x + 1$ and $x^2 + 1$, such that we can apply the results from ρ_* and τ_* to determine the orbits of size three and two.

- (i) For $\ell = 2, 3$ this is a direct computation.
- (ii) For $\ell \equiv 1 \pmod{12}$, we have that Φ_{12} splits completely and for some symplectic basis, the automorphism is given by $\mathrm{diag}(\zeta_{12}, \zeta_{12}^5, \zeta_{12}^{11}, \zeta_{12}^7) \in \mathrm{Sp}_4(\mathbb{F}_\ell)$. This yields 4 invariant maximal isotropic subspaces by [Lemma 3](#).
- (iii) For $\ell \equiv 5, 7, 11 \pmod{12}$, we have that Φ_{12} factors as respectively $(x^2 + ix - 1)(x^2 - ix - 1)$, $(x^2 + \mu_3)(x^2 + \mu_3^2)$ and $(x^2 + \sqrt{3}x + 1)(x^2 - \sqrt{3}x + 1)$, resulting in respective matrices

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & -i & 0 & 0 \\ 0 & 0 & i & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ \mu_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\mu_3 & 0 \end{pmatrix}, \begin{pmatrix} \sqrt{3} & 0 & -1 & 0 \\ 0 & -\sqrt{3} & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \in \mathrm{Sp}_4(\mathbb{F}_\ell).$$

None of these cases yield similar 2×2 blocks, so by [Lemma 4](#), only the subspaces corresponding to the two blocks can be potential invariant maximal isotropic subspaces. This is the case for $\ell \equiv 5, 7 \pmod{12}$ but for $\ell \equiv 11 \pmod{12}$ they are not isotropic.

□

This yields the following result.

Theorem 13. *Let $E_0 \times E_0$ be a product of elliptic curves over a field of characteristic $p > 5$, with $\mathrm{RA}(E_0 \times E_0) = \mathbb{Z}/6\mathbb{Z} \times \mathcal{S}_3$, and $\ell \neq p$ a prime. Then the number of (ℓ, ℓ) -level surfaces $(E_0 \times E_0, G)$ is equal to 3 if $\ell = 2$, equal to 5 if $\ell = 3$, and for $\ell > 3$ it is*

$$\frac{1}{36} \left(\ell^3 + 5\ell^2 + 24\ell + 59 + (8\ell + 38) \left(\frac{\ell}{3} \right) + 9 \left(\frac{-1}{\ell} \right) \right).$$

Proof. All automorphisms of $E_0 \times E_0$ are given by $(P, Q) \mapsto ((-\zeta)^a(P), (-\zeta)^b(Q))$ or $(P, Q) \mapsto ((-\zeta)^c(P), (-\zeta)^d(Q))$, where $b, d \in \{0, 1, 2\}$ and $a, c \in \{0, 1, 2, 3, 4, 5\}$ cover all options up to equivalence. One can verify that this group contains

- one identity element id ;
- seven elements of order two: four with minimal polynomial $x^2 - 1$ acting as κ from [Proposition 8](#) and three elements with minimal polynomial $x^2 + 1$ acting as τ_* from [Proposition 2](#);

- eight elements of order three: two with minimal polynomial $x^2 + x + 1$ and two with minimal polynomial $(x + 1)(x^2 + x + 1)$ which act as ρ_* from [Proposition 3](#), four with minimal polynomial $(x - 1)(x^2 + x + 1)$ which act as $[1] \times \zeta^{(3)}$ from [Proposition 10](#);
- twenty elements of order six: two with minimal polynomial $x^2 - x + 1$ and two with minimal polynomial $(x - 1)(x^2 - x - 1)$ which act as $[1] \times \zeta^{(3)}$ from [Proposition 10](#), ten with minimal polynomial $(x^2 + x + 1)(x^2 - x + 1)$ which act as $\zeta_*^{(6)}$ from [Proposition 4](#), six with minimal polynomial $x^4 - x^2 + 1$ which act as $\chi \circ ([1] \times -\zeta^{(3)})$ from [Proposition 12](#).

We thus have that the number of (ℓ, ℓ) -level surfaces is given by

$$\frac{1}{36} \cdot \left(|X^{\text{id}}| + 4|X^\kappa| + 3|X^{\tau*}| + 4|X^{\rho*}| + 8|X^{[1] \times \zeta^{(3)}}| + 10|X^{\zeta_*^{(6)}}| + 6|X^{\chi \circ ([1] \times -\zeta^{(3)})}| \right).$$

□

C Representation of (ℓ, ℓ) -isogenies

In this section, we give a compact description of all maximal isotropic (ℓ, ℓ) -subgroups, including a unique representation for such a subgroup, given a symplectic basis of $J[\ell]$. This representation is very similar to one given by Kunzweiler, Ti, and Weitkämper [\[KTW22\]](#), although we adapt their representation to our purpose of computing (ℓ, ℓ) -isogenies, for which we require a specific representation of maximal isotropic subgroups. This representation has two advantages: on the theoretical side, they simplify our later exposition of the action of automorphisms on such subgroups. On the practical side, they allow us to quickly compute all such subgroups and speed up the computation of orbits under the automorphism.

The echelon form. We assume a symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$ of $J[\ell]$, i.e., a basis such that

$$e_\ell(D_1, D_2) = 1, \quad e_\ell(D'_1, D'_2) = 1, \quad e_\ell(D_i, D'_i) = \mu_\ell,$$

where e_ℓ is the ℓ -Weil pairing and μ_ℓ some fixed ℓ -th root of unity. Any maximal isotropic (ℓ, ℓ) -subgroup G is then generated by two elements

$$K_1 = [a_1]D_1 + [a_2]D_2 + [a_3]D'_1 + [a_4]D'_2, \quad K_2 = [b_1]D_1 + [b_2]D_2 + [b_3]D'_1 + [b_4]D'_2$$

such that $e_\ell(K_1, K_2) = 1$. Written additively, these coefficients must satisfy the following equation in order for G to be maximal isotropic:

$$\Delta(G) := a_1 \cdot b_3 + a_2 \cdot b_4 - a_3 \cdot b_1 - a_4 \cdot b_2 \equiv 0 \pmod{\ell}. \quad (3)$$

By representing (the generators K_1 and K_2 of) a maximal isotropic subgroup G as a matrix

$$M_G = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \end{pmatrix}, \quad \text{with } \Delta(G) \equiv 0 \pmod{\ell},$$

we find a simple way to compute all $\ell^3 + \ell^2 + \ell + 1$ maximal isotropic subgroups, together with a canonical representation, by simple row reductions on M_G . We refer to such a row-reduced representation $\overline{M}_G$ as the *echelon form* of G , with respect to a symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$.

When a_1 is non-zero. If a_1 is non-zero, we may normalize the first row, and clear the second row with the first row. Depending on the number of leading zeroes in the second row, this gives us three options after normalizing to the first non-zero value.

- (i) If the resulting b_2 is non-zero, we get

$$\overline{M}_G = \begin{pmatrix} 1 & 0 & a_3 & a_4 \\ 0 & 1 & b_3 & b_4 \end{pmatrix}, \quad \text{with } \Delta(G) = b_3 - a_4 \equiv 0 \pmod{\ell}. \quad (4)$$

This gives us ℓ^3 subgroups by choosing $a_3, a_4, b_4 \in \mathbb{Z}/\ell\mathbb{Z}$, and are required to set $b_3 = a_4$.

- (ii) If the resulting b_2 is zero, but b_3 is non-zero, we get

$$\overline{M}_G = \begin{pmatrix} 1 & a_2 & 0 & a_4 \\ 0 & 0 & 1 & b_4 \end{pmatrix}, \quad \text{with } \Delta(G) = 1 + a_2 \cdot b_4 \equiv 0 \pmod{\ell}. \quad (5)$$

This gives us $\ell \cdot (\ell - 1)$ options by choosing $a_2 \in (\mathbb{Z}/\ell\mathbb{Z})^\times$ and $a_4 \in \mathbb{Z}/\ell\mathbb{Z}$, and are required to set $b_4 = -a_2^{-1}$.

- (iii) If the resulting b_2 and b_3 are zero, but b_4 is non-zero, we get

$$\overline{M}_G = \begin{pmatrix} 1 & a_2 & a_3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{with } \Delta(G) = a_2 \equiv 0 \pmod{\ell}. \quad (6)$$

This gives us ℓ options, as we may only choose $a_3 \in \mathbb{Z}/\ell\mathbb{Z}$ freely.

When a_1 is zero. We may do similar derivations assuming both a_1 and b_1 are zero, as we simply swap rows. This gives us three other options for canonical representations.

- (i) If b_3 is non-zero after clearing row two, we get

$$\overline{M}_G = \begin{pmatrix} 0 & 1 & 0 & a_4 \\ 0 & 0 & 1 & b_4 \end{pmatrix}, \quad \text{with } \Delta(G) = b_4 \equiv 0 \pmod{\ell}. \quad (7)$$

This gives us another ℓ options, as we may only choose $a_4 \in \mathbb{Z}/\ell\mathbb{Z}$ freely.

- (ii) If b_3 is zero after clearing row two, we get

$$\overline{M}_G = \begin{pmatrix} 0 & 1 & a_3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{with } \Delta(G) = 1 \equiv 0 \pmod{\ell}. \quad (8)$$

As the requirement on $\Delta(G)$ is impossible, none of these subgroups are maximal isotropic.

- (iii) Lastly, the only remaining option is the maximal isotropic subgroup given by

$$\overline{M}_G = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{with } \Delta(G) = 0 \pmod{\ell}. \quad (9)$$

The action of α on the echelon form. When α is a (reduced) automorphism of a curve C , with α_* the associated automorphism on its Jacobian J , we may write the action of α_* on the symplectic basis $\langle D_1, D_2, D'_1, D'_2 \rangle$ as some matrix $M_\alpha \in \mathbb{F}_\ell^{4 \times 4}$. We then easily compute the action of α_* on G by $\overline{M_{\alpha_*(G)}} = \text{Echelon}(M_\alpha \cdot M_G^T)$.

The echelon form allows us to deduce results on the size of orbits of subgroups under automorphisms, by identifying those echelon forms with orbits of certain orders. In particular, for computational purposes, the echelon form allows us to easily compute all maximal isotropic subgroups, given a symplectic basis. Furthermore, as the echelon form $\overline{M_G}$ is a unique representation of G , this allows us to easily verify equality of subgroups G and G' by comparing $\overline{M_G} = \overline{M_{G'}}$, simplifying an otherwise tedious comparison.

C.1 Good extensions for the echelon form

The categorization using echelon form makes it simple to compute the good extensions to a given isogeny $\varphi : J' \rightarrow J$ which we may represent as (J, G) where $G = \ker \hat{\varphi}$.

Lemma 6. *Let (J, G) be a level structure of a p.p. abelian surface with $G = \langle T_1, T_2 \rangle$ for $T_1, T_2 \in J[\ell]$. Let $S_1, S_2 \in J[\ell]$ be such that S_1, S_2, T_1, T_2 is a symplectic basis. Then the ℓ^3 good extensions of (J, G) are given by the kernels $\langle K_1, K_2 \rangle$, where*

$$K_1 = S_1 + [a]T_1 + [b]T_2, \quad K_2 = S_2 + [b]T_1 + [c]T_2, \quad a, b, c \in \mathbb{Z}/\ell\mathbb{Z}.$$

Proof. This follows directly from [Eq. \(4\)](#). $\square$

It is similarly straightforward to compute the good extensions on products of elliptic curves.

Lemma 7. *For an elliptic product $(E \times E', G)$ with (ℓ, ℓ) -level structure, with associated isogeny $\varphi : E \times E' \rightarrow (E \times E')/G$, we have:*

- if φ is a diagonal isogeny, then there are ℓ^2 diagonal kernels G' and $\ell^3 - \ell^2$ gluing kernels G' with $G \cap G' = 0$;
- if φ is a gluing isogeny, then there are $\ell^2 + \ell$ diagonal kernels G' and $\ell^3 - \ell^2 - \ell$ gluing kernels G' with $G \cap G' = 0$.

Proof. When φ is diagonal, we may choose a basis $(Q, \infty), (\infty, Q')$ for G , which we may extend to a symplectic basis $\langle (P, \infty), (\infty, P'), (Q, \infty), (\infty, Q') \rangle$ for $(E \times E')[\ell]$. Similar to [Lemma 6](#), we can associate a good extension to a triple $(a, b, c) \in (\mathbb{Z}/\ell\mathbb{Z})^3$. Diagonal kernels are those with $b = 0$.

When φ is a gluing, we may assume $\ell^2 - 1$ points (K_1, K_2) of order ℓ in its kernel. Each of these overlaps with the diagonal isogeny generated by $(K_1, \infty), (\infty, K_2)$. Dividing by the projective factor $\ell - 1$ to remove scalars, this gives us $\ell + 1$ bad diagonal kernels out of $(\ell + 1)^2$ total diagonal kernels, leaving $\ell^2 + \ell$ good ones. $\square$